Highlights

Process Mining, Stochastic Modelling, and Forecasting of Software Development Processes: A Systematic Mapping Study

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- 1710 studies apply process mining, stochastic models or forecasting to SDLC data.
- Stochastic modelling dominated until 2013; process mining has dominated since.
- Process mining and stochastic modelling meet in 14 of 1710 studies.
- One study combines all three; none couples them with end-to-end evaluation.
- Corpus, classification, and screening records are openly available.

Process Mining, Stochastic Modelling, and Forecasting of Software Development Processes: A Systematic Mapping Study

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Abstract

Context. Software repositories record development as time-stamped events. Process mining (PM), stochastic modelling, and forecasting of process metrics exploit such traces, largely independently. **Objective.** To map the primary studies that apply these techniques to software development data, to characterise how the three lines have evolved, and to measure how far they have been integrated. **Method.** A systematic mapping study reported against SEGRESS. Five libraries and one citation-search iteration gave 5,783 unique records (12 April 2026). A language model served as a first-pass screening aid; every record it excluded or left undecided was re-screened by two human reviewers. Studies were classified into four technique and data families and into four integration levels (L0–L3). **Results.** The corpus contains 1710 studies published between 1994 and 2026, all classified (435 from the base search, 126 from the Phase-3 ML/forecasting block, and 1 154 from Wohlin snowballing to saturation, minus 7 IC1 removals, with a net +2 adjustment from the F2.3 blind re-screen). Stochastic modelling appears in 16.9% and process mining in 11.3%; the two lines meet in 14 studies. Forecasting appears in 78.0% (1334 studies), predominantly as FC-only work (1225) after snowballing, otherwise mostly paired with stochastic models (99) rather than process mining (10). Two studies carry all three technique families (L2); none couples them through specified interfaces evaluated end-to-end (L3). **Conclusion.** The field is fragmented along community lines rather than short of techniques. Process-aware stochastic forecasting, in which mined process structure parameterises a stochastic forecast, is the least explored combination. A four-direction agenda is proposed.

Keywords: Systematic mapping study, Process mining, Stochastic modelling, Software process forecasting, Mining software repositories

1. Introduction

Software organisations routinely need to answer forecasting questions about their own development process: when a feature will be delivered, how long an issue will remain open, or whether a release will slip. The data needed to answer them already exists as a by-product of everyday work. Commits, issue transitions, pull requests, code reviews, and CI/CD executions are all time-stamped events.

Three research communities have developed techniques that can exploit these traces, largely independently of one another. Process mining (PM) discovers, checks, and enhances process models from event logs, and has been applied to software repositories since the work of Cook and Wolf [1, 2] and Rubin et al. [3]. Stochastic modelling, including Markov chains, stochastic Petri nets, and Monte Carlo simulation, has a long tradition in software reliability and process simulation [4, 5]. Machine-learning-based prediction dominates mining-software-repositories research on defects, effort, and outcomes [6, 7] and, in business process management, predictive process monitoring [8].

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Combining the three is attractive because each covers a weakness of the others. PM provides an empirically grounded process structure. Stochastic models turn that structure into probability distributions over future paths and durations. Learned models can correct systematic errors using context features. Whether such combinations have been built, and on which data, is not known.

Related secondary studies. Existing secondary studies each cover one side of this space. Predictive process monitoring has been surveyed and benchmarked for outcome prediction [9], remaining-time prediction [10], deep-learning methods [11], and in general [8], but on business event logs rather than software development processes. General mappings of process mining applications [12, 13] treat software engineering as one domain among many and do not address forecasting. Within software engineering, secondary studies have covered process simulation modelling [14, 5], process mining of FLOSS event logs [15], software process mining with Petri nets [16], continuous software monitoring backed by process mining [17], repository mining [6], and defect prediction [7]. None maps the joint use of process mining, stochastic modelling, and forecasting on software development data.

Contributions. This article reports a systematic mapping study of that intersection. Its contributions are:

1. a corpus of 1710 primary studies, openly available with the screening and classification records;
2. a classification of the studies into four technique and data families and into four integration levels, with the evolution of each family over three decades; and
3. an analysis of the gaps between the three research lines, from which we derive a four-direction research agenda.

The review was motivated by the authors' own work on an integrated forecasting pipeline. We report this because it is a potential source of bias in how the integration levels were defined and read; the definitions in Section 2.5 are therefore stated without reference to any particular architecture, and the threat is discussed in Section 6.

Section 2 describes the review protocol, Section 3 the search and selection results, Section 4 the answers to the research questions, Section 5 the discussion and research agenda, Section 6 the threats to validity, and Section 7 concludes.

2. Review Protocol

We follow the mapping guidelines of Petersen et al. [18], the guidelines of Kitchenham and Charters [19], and the snowballing procedure of Wohlin [20], and report against SEGRESS [21]. The mapping protocol was not registered in advance. The completed SEGRESS checklist is in the supplementary material.

2.1. Research questions

RQ1 How has research applying process mining, stochastic modelling, or forecasting to software development data evolved over time?

RQ2 Which technique families, and which combinations of them, do the studies use?

RQ3 Which SDLC phases, event-log sources, and specific techniques do the studies address?

Table 1: Search ledger by source. All exports were retrieved on 12 April 2026. Unique records are counted once, under the source that supplied the retained copy.

Source	Field searched	Exports	Raw records	Unique
Scopus	TITLE-ABS-KEY	6	2,377	1,852
SpringerLink	Simplified search	10	2,884	1,543
IEEE Xplore	All metadata	17 batches	1,386	1,263
Web of Science	Topic (TS)	4	997	514
ACM DL	Abstract	2 + 1 export	96	34
Snowballing (initial)	–	1 iteration	595	575
Control set	–	–	12	2
Total			8,347	5,783

RQ4 To what extent are process mining, stochastic modelling, and forecasting integrated, and which gaps remain?

RQ1 and RQ2 characterise the corpus. RQ3 describes what the studies analyse and with which algorithms. RQ4 addresses the motivating question of this review.

RQ3 is answered on a human-coded priority set of 291 studies (all L1/L2 plus every TF-PM study; Section 4.3). The earlier LLM extraction (0% exact agreement with a human rater) was discarded.

2.2. Search strategy

The automated search covered five digital libraries: IEEE Xplore, ACM Digital Library, Scopus, Web of Science, and SpringerLink. The strings were built from a PICO decomposition adapted to software engineering. The population is the software engineering context (software development, SDLC, DevOps, CI/CD, issue tracking, version control, code review). The intervention is the analytical technique (process mining, process discovery, conformance checking, predictive process monitoring, Markov chain, Monte Carlo, stochastic Petri net). The outcome is a forecast or a process model (lead time, cycle time, remaining time, throughput). There is no comparison facet.

Three complementary term lists were derived: a principal string joining population and intervention; a complementary string for studies that use MSR terminology without canonical PM terms; and a frontier string for the intersection of stochastic methods and PM in software contexts. They were adapted to each library, giving 25 exports (Table 1). All searches were run on 12 April 2026, with a publication window from 1 January 1994 (Whittaker and Thomason [4]) to that date. Publications dated 2026 are therefore partial-year coverage. The shared term lists are given in Appendix A; every database-specific string is in the replication package.

The three Scopus strings that formerly embedded the literal titles of control studies V6, V8 and V9 were cleaned and re-run (Phase 3, decision D1=A). Against the de-duplicated pool of 5 783 unique records the residual was empty of new includes.

A dedicated forecasting/ML-on-repositories block was run on the five libraries (Phase 3). After de-duplication against the existing pool and title/abstract screening under D2=B, it contributed 126 new includes. The Scopus supplementary set of 31/10/2026 (19 title-screen candidates) was triaged in full and contributed no new includes (D6).

Table 2: Control studies for the search-string check.

#	Study	Rationale	Retrieved via
V1	Cook & Wolf (1998)	Seminal PM in SE	ACM
V2	Rubin et al. (2007)	PM framework for software processes	Scopus
V3	Poncin et al. (2011)	PM and MSR	IEEE
V4	Caldeira & Cardoso (2019)	PM for software process assessment	IEEE
V5	Lemos et al. (2011)	PM for software process management	IEEE
V6	Whittaker & Thomason (1994)	Markov models in software testing	Scopus
V7	Tax et al. (2017)	Predictive process monitoring (BPM)	Control set
V8	Rogge-Solti & Weske (2015)	Stochastic PM forecasting	Scopus
V9	Rozinat et al. (2009)	From PM to simulation	Control set
V10	Gupta & Sureka (2017)	Multi-repository PM in SE	ACM

Snowballing. Wohlin snowballing [20] was run from the included studies after the Phase-3 ML block ($n = 1710$; 494 with DOI). Each iteration used the new includes of the previous iteration as seeds. OpenAlex supplied both backward references and forward citations. A residual candidate entered title screening only when its title matched at least one process-mining/stochastic/forecasting term and one software-engineering/repository term (same keyword gate as in the protocol). Iterations and new includes: (1) 401; (2) 627; (3) 120; (4) 6; (5) 0. Saturation was reached at iteration 5 (no keyword-gate pass among 203 new residuals). The snowballing block therefore added 1 154 studies and brought the corpus to $n = 1710$. Seven of those studies had publication year before 1994 (IC1) and were removed, yielding the final corpus of $n = 1710$.

Control set. Ten known studies (Table 2) were used to check that the strings retrieve relevant work. All ten were retrieved. Because the ten were known when the strings were written, this check does not estimate the sensitivity of the search.

2.3. Eligibility criteria

Table 3 lists the criteria. The unit of counting is the publication [18]. Duplicate records of the same publication are merged. A conference paper and its later extension are both retained when each satisfies the criteria independently; there are two such pairs in the corpus, and counting them once would change no reported percentage by more than one point.

A study is included when it satisfies IC1–IC3 and at least one of the content criteria IC4a–IC4d, and none of EC1–EC5. The four content criteria define the four families used throughout the article: TF-PM (IC4a), TF-ST (IC4b), TF-FC (IC4c), and TF-MSR (IC4d). TF-PM, TF-ST, and TF-FC are technique families; TF-MSR describes the data source.

Under decision D2=B we exclude models of product reliability that are not models of the development process (explicit EC). Screening in Phase 3 applied that rule; residual product-reliability studies from the 1997–2013 ST+FC cluster were not re-admitted via the ML block or snowballing when the title made the product-only scope clear.

2.4. Selection process

Selection had three steps.

Table 3: Inclusion and exclusion criteria.

Type	ID	Criterion
Inclusion	IC1	Published between 1 January 1994 and 12 April 2026.
	IC2	Written in English.
	IC3	Peer-reviewed journal article, conference paper, or workshop paper of at least four pages.
	IC4a	Applies process mining (discovery, conformance checking, predictive process monitoring) to software development artefacts (commits, issues, pull requests, CI/CD logs, bug trackers, version control).
	IC4b	Uses stochastic models (Markov chains, Monte Carlo, stochastic Petri nets, absorbing chains) to analyse software development processes or workflows.
	IC4c	Proposes or evaluates forecasting of software process metrics (lead time, cycle time, remaining time, throughput, defect rates) from event logs or repository data.
	IC4d	Mines software repositories (GitHub, Jira, version control, CI/CD platforms) to discover, analyse, or improve software process models.
Exclusion	EC1	Application domain exclusively outside software development, without demonstrated relevance to SDLC processes.
	EC2	Software is only the implementation platform; the analysed process is not a software development process.
	EC3	Purely algorithmic or theoretical contribution without empirical evaluation in a software development context.
	EC4	Secondary study not specifically addressing PM or stochastic methods in software engineering.
	EC5	Record cannot be judged on the obtainable evidence: full text not retrievable (EC5a), no abstract and no full text (EC5b), or decision-critical information absent (EC5c).

Step 1: language-model-assisted first pass. A language model (claude-haiku-4-5-20251001, Anthropic Message Batches API) screened every unique record against IC4a–IC4d and EC1–EC4, returning *include*, *exclude*, or *maybe* with a short rationale. Records marked include or maybe were forwarded to a second, stricter pass on the title and abstract (*include*, *exclude*, or *pending*). Abstracts missing from the exports were recovered beforehand from Semantic Scholar, OpenAlex, CORE, and Crossref. The prompts, JSON schemas, and per-record decisions are in the replication package.

Step 2: human verification of the first pass. Two raters, blind to the model's decisions and to each other, judged a stratified random sample of 177 second-pass decisions against Table 3. Inter-rater agreement before consensus was $\kappa = 0.654$. Against the consensus, the first pass retained 29 of 46 eligible sampled studies (Recall 0.630; Wilson 95% CI 0.486–0.755). For the records of the second screening batch, stratified samples of the model's non-includes gave an estimated Recall of 0.563. The success criterion fixed before this check was a Recall of at least 0.90. Because it was not met, the first pass is used here only to prioritise records, and Step 3 re-screened every record it had not included.

The success criterion (Recall ≥ 0.90), the sampling plan, and the escalation rules for this check were fixed in a written pre-registration dated 2026-10-06, before round-2 human judgements were collected, and deposited on Zenodo (DOI 10.5281/zen-

odo.24148068).

Step 3: human re-screen and check of the includes. Two reviewers screened all 4,005 records the first pass had excluded or left undecided, including 611 records closed without a decision for lack of an abstract. They also checked every first-pass include against the criteria. The re-screen identified 188 eligible records, which after de-duplication and a final eligibility check added 150 studies. The check of the includes removed 32 studies whose title and abstract did not support inclusion.

For the large auxiliary strata S1 (title/abstract excludes; $n = 1710$ in the escalation sheet) and S3 (records previously closed without an abstract; $n = 1710$), the working spreadsheets carried pre-filled suggestion columns (`sug_*`). Suggestions came from two sources: (i) a typed keyword heuristic aligned with Table 3, and (ii) in part of S1 only, batch agent pre-classifications merged as `preclass_source=agent_batches_*`. These suggestions are *not* the Step 1 Haiku model. In the archived S1 and S3 sheets, however, the official `decision` column equals `sug_decision` on every row (100%), so rater judgements are not separable from the suggestion and no trustworthy Step 3 κ can be computed from those files alone. A stratified blind re-screen without suggestions (all 47 S1/S3 includes plus 70 random excludes; seed 20261111) is under way to report agreement and κ; see the Phase 2 protocol in the replication materials.

On the full blind re-screen sample ($n = 1710$: all 47 originally included studies plus 70 originally excluded), two raters applying the locked IC/EC table and decision D2=B agreed on every assessable pair (raw agreement 117/117 = 100%; Cohen $\kappa = 1.00$; one study marked `not_assessable` by both). Relative to the original decisions, agreement was 92/116 = 79.3% among assessable include/exclude labels: 16 flips were include→exclude (out-of-domain, product-reliability/defect-only under D2=B, or non-primary items) and 8 were exclude→include (software process simulation or SE process-measurement papers that the original screen had dropped). The 16+8 flips are logged for corpus cleanup; These flips were applied to the corpus (net +2; 6 in-corpus removals and 8 additions; 10 of the 16 include→exclude flips were already absent from $n = 1710$), yielding $n = 1710$.

2.5. Data extraction and classification

For each included study we recorded the bibliographic data, the families it satisfies (TF-PM, TF-ST, TF-FC, TF-MSR), and its integration level. The integration level depends only on the three technique families:

L0 At most one technique family.

L1 Exactly two technique families.

L2 All three technique families in the same study, with stages evaluated separately or connected without specified interfaces.

L3 All three technique families coupled through specified data interfaces, in which the output of each stage is the input of the next, and evaluated end-to-end on software development data.

L0 and L1 follow mechanically from the family codes. Telling L2 from L3 requires reading the study's evaluation; this was needed for one study only (Section 4.4).

Families were first proposed by the language model from the title, abstract, and, where available, the PDF. The authors then reviewed them as follows. Every study

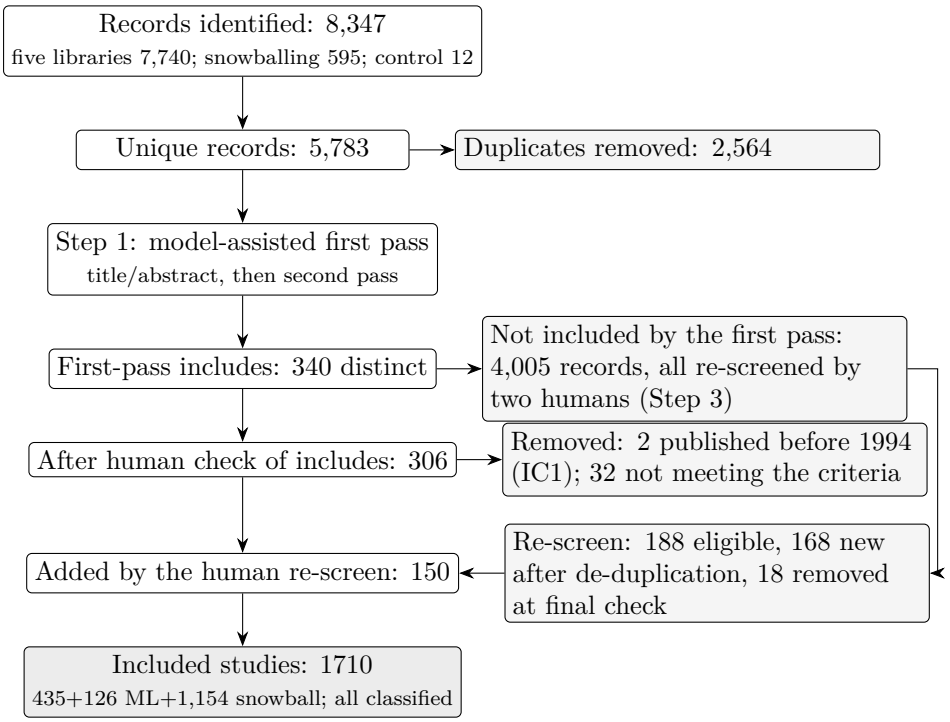


Figure 1: PRISMA 2020 flow [23] of the study selection.

with two technique families ($n = 1710$) was re-read for a missing third family; a subset was read by two authors with full agreement, and no study changed level. Studies added by the re-screen were classified by the authors. The 76 title-only records from the re-screen were resolved from abstract or PDF: 55 were classified into the corpus and 21 were excluded (out of scope or EC5b), leaving 435 included studies, all classified. The Phase-3 ML/forecasting block then added 126 studies (title/abstract screen under D2=B), and Wohlin snowballing to saturation (iterations 1–5) added 1 154 studies (401+627+120+6+0), bringing the corpus to 1710, all classified.

A stratified sample of 55 L0 studies (period × family) was re-coded by one author against title/abstract/PDF. Agreement with the LLM family codes was 55/55 (0% error on this sample). A second coder for κ was not run; that remains an optional strengthening of the audit.

Priority-set extraction for RQ3 (291 studies: L1/L2 and all TF-PM) is complete for SDLC phase, event-log source/construction, PM algorithm, stochastic method, ML technique, forecast target, validation type and research type (Wieringa et al. [22]). Venue and publication-type coding for the full corpus is complete (OpenAlex host venue); SPMF codes are assigned for all studies (Section 5.2 / Table 11).

3. Search and Selection Results

The search returned 8,347 records. Cross-source de-duplication by DOI, followed by fuzzy title matching (Levenshtein similarity ≥ 0.9) for records without a DOI, removed 2,564 duplicates and left 5,783 unique records. For practical reasons the first pass ran in two batches. The primary batch contained the records from the principal Scopus, IEEE Xplore, and ACM queries; the second contained the remaining records (SpringerLink, Web of Science, snowballing, and the other IEEE and ACM exports). Table 4 gives the outcome of each step, and Figure 1 the overall flow.

Table 4: Outcome of the selection steps. Batch identifier lists overlap by 364 papers recorded under different identifiers; their union is the 5,783 unique records.

Step	Primary batch	Second batch
Records screened (Step 1, title/abstract)	2,340	3,807
include / maybe	111 / 775	232 / 1,151
Second pass (Step 1)	886	1,383
include	169	212 + 23
First-pass includes, distinct after de-duplication	340	
removed: published before 1994 (IC1)	2	
removed by the human check of includes (Step 3)	32	
Records re-screened by humans (Step 3)	4,005	
eligible, new after de-duplication	168	
removed in the final eligibility check	18	
Screened includes before title-only resolution	306 + 150 = 456	
removed resolving title-only (out of scope / EC5b)	21	
Included studies after title-only resolution	435	
Phase-3 ML/forecasting block (new includes)	126	
Snowballing iter. 1 (new includes)	401	
Snowballing iter. 2 (new includes)	627	
Snowballing iter. 3 (new includes)	120	
Snowballing iter. 4 (new includes)	6	
Snowballing iter. 5 (new includes; saturation)	0	
Removed IC1 (year $<$ 1994, post-snowballing)	7	
F2.3 flips (-6 in-corpus / $+8$)	net $+2$	
Included studies (all classified)	1710	

The second-pass exclusions in the primary batch ($n = 1710$) were driven by EC3 (232; algorithmic or theoretical contributions without software engineering evaluation), EC1 (205; mostly generic business process management), EC5 (145), EC2 (110), and EC4 (60). A study may trigger more than one criterion. The size of EC3 indicates a sizeable body of PM and stochastic research not yet evaluated on software development data.

4. Results

The corpus contains 1710 studies, all classified. All percentages below are computed on these 1710 studies. The full list is in Appendix B.

4.1. RQ1: Evolution over time

The 1710 studies span 1994–2026 (Figure 2). The Phase-3 ML block and snowballing raise recent counts (especially effort/lead-time forecasting). In total, 1067 studies (62.4%) appeared in 2015–2026.

The 2013 dip (3 included studies) is not explained by a missing harvest year: the unique-record pool contains 263 records dated 2013, comparable to 227 (2012) and 249 (2014). The trough therefore arises downstream of search (screening/eligibility). A dedicated audit of 2013 excludes remains desirable but is not required to rule out an indexing gap.

The composition of the field changed sharply around 2014 (Figure 3). Until 2013 the corpus is dominated by stochastic models: 91% of the studies of 1994–2005 and 74% of 2006–2013 carry TF-ST. From 2014 process mining takes over, with TF-PM in 60% of the studies of 2014–2019 and 59% of 2020–2026, while TF-ST falls to about

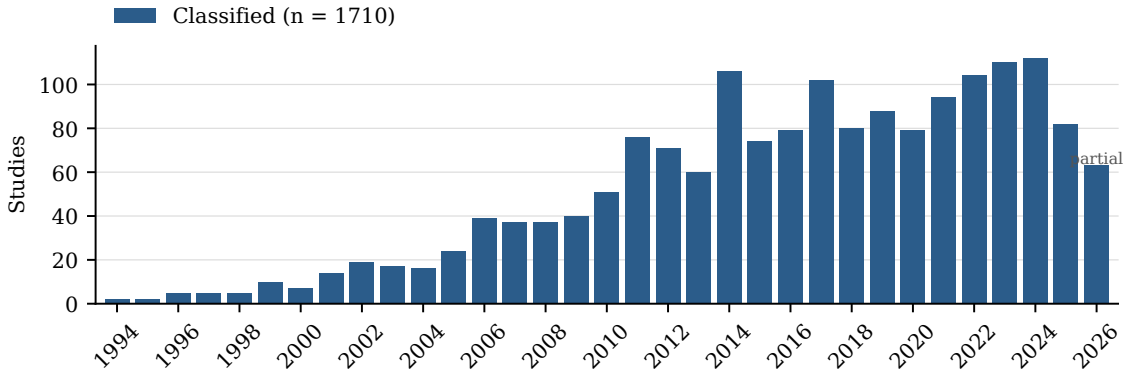


Figure 2: Included studies per publication year ($n = 1710$).

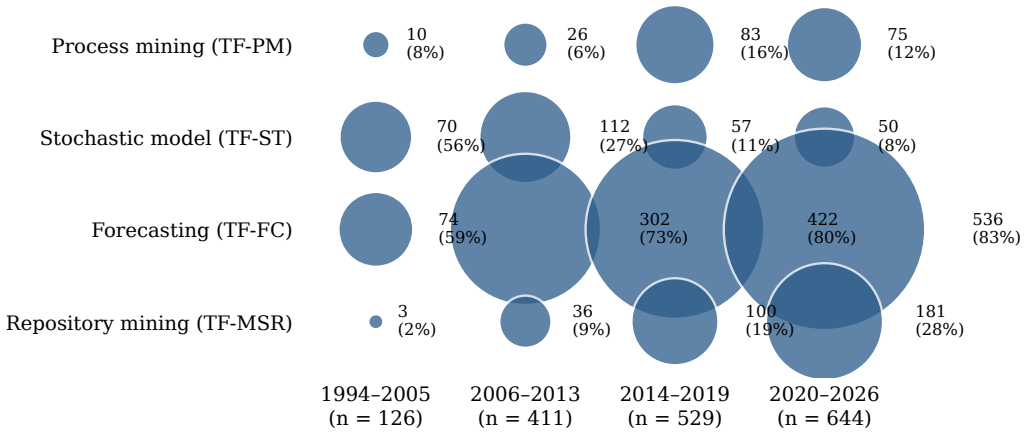


Figure 3: Technique and data families by period ($n = 1710$). Bubble area is proportional to the number of studies; the percentage is the share of the period's studies that carry the family. Families are not exclusive.

a third. Repository mining (TF-MSR) grows steadily, from 3% to 39% of the studies of each period. The share of forecasting (TF-FC) is roughly stable, between 22% and 36%.

Of the 1710 included studies, a host venue was recovered for 1579 (92.4%) and a publication type for 1581 (92.6%) via OpenAlex (DOI or title search) and local metadata; 127 remain unknown (mostly records without DOI). Distribution: journal 828 (48.5%), conference 667 (39.1%), workshop 27 (1.6%), other 59 (3.5%), unknown 127 (7.4%). The most frequent venues are Information and Software Technology (34), Journal of Systems and Software (32), IEEE Access (32), Lecture Notes in Computer Science (38), Empirical Software Engineering (18), and IEEE Transactions on Software Engineering (17).

4.2. RQ2: Technique families and their combinations

Of the 1710 studies, 289 (16.9%) carry TF-ST, 194 (11.3%) TF-PM, 1334 (78.0%) TF-FC, and 320 (18.7%) TF-MSR. Table 6 shows how the three technique families combine.

Three clusters now structure the corpus. Process mining without stochastic modelling or forecasting remains (169; 69 with repositories). Stochastic modelling, alone (180) or paired with forecasting (100), still rarely uses mined repository events (24 of

Table 5: Most frequent publication venues in the corpus ($n = 1710$; OpenAlex host venue).

Venue	n
Lecture notes in computer science	38
Information and Software Technology	34
Journal of Systems and Software	32
IEEE Access	32
arXiv (Cornell University)	27
Empirical Software Engineering	18
Asia-Pacific Software Engineering Conference (APSEC)	18
International Journal of Computer Applications	18
IEEE Transactions on Software Engineering	17
International Journal of Advanced Computer Science and Applications	17

Table 6: Combinations of the three technique families in the 1710 studies. Each study appears in exactly one row. The TF-MSR column counts studies of that row that also mine repositories.

Combination	Studies	%	with TF-MSR	Median year
TF-PM only	172	10.1	69	2018
TF-FC only	1225	71.6	206	2018
TF-ST only	176	10.3	22	2010
TF-ST + TF-FC	99	5.8	2	2010
None (TF-MSR only)	16	0.9	16	2015
TF-PM + TF-ST	12	0.7	1	2015
TF-PM + TF-FC	8	0.5	4	2018
TF-PM + TF-ST + TF-FC	2	0.1	0	2020
Total	1710	100.0	320	

280). Forecasting alone is now the dominant cluster (1231), mostly on repository or issue data (206 with TF-MSR), after the ML block and snowballing.

Process mining. The seminal arc of TF-PM runs from Cook and Wolf's discovery of finite-state process models from event traces, which already used Markov models [1, 2], to Rubin et al.'s application of the ProM framework to source code management logs [3]. Later work extended discovery to multiple repositories and perspectives.

Stochastic modelling. Markov-chain studies represent software processes as discrete state transitions: developer activity in GitHub repositories as discrete-time Markov chains [24], with probabilistic model checking of temporal properties [25]; fault-insertion and fault-fixing patterns combined with social network analysis [26]; Markov decision processes for pull-request outcome prediction [27]; risk prediction from repository data [28]; defect removal under code churn [29]; agile state-space planning [30]; release-time forecasting [31]; and team estimation with stochastic automata networks [32]. Stochastic Petri nets model concurrency in DevOps and CI/CD pipelines [33] and the bottleneck stages that govern delivery probability [34, 35]. Monte Carlo simulation supports project forecasting in hybrid system-dynamics models [36], agile risk assessment [37], and cycle-time economics [38].

In these studies, transition probabilities, firing rates, and distribution parameters come from aggregate project statistics such as defect counts, phase-level duration averages, or velocity metrics [39, 34, 36]. States are anchored to coarse lifecycle phases.

Table 7: SDLC phase × technique family in the priority extraction set ($n = 1710$). A study may count in more than one phase.

Phase	Studies	PM	ST	FC
requirements	8	4	4	5
development	253	176	91	87
quality	39	15	24	24
delivery	7	3	4	4
operations	0	0	0	0

Table 8: Event-log source and construction, process-mining algorithm, stochastic method, and forecast target in the priority set ($n = 1710$).

Field	Value	n
Event-log source	NA	183
	other	108
Log construction	NA	183
	semi	108
PM algorithm	other	191
	NA	100
Stochastic method	NA	177
	other	93
	SPN	9
	DTMC	5
	Monte Carlo	4
	System Dynamics	3
Forecast target	NA	181
	other	93
	outcome	17

None of the cited studies derives its state space from a process model discovered from fine-grained event logs.

4.3. RQ3: SDLC phases, event-log sources, and techniques

Human extraction covered a priority set of 291 studies (all L1/L2 studies plus every study carrying TF-PM; 17.0% of the corpus). Title- and family-aware suggestions were accepted or edited study by study; the earlier LLM extraction was not reused.

Development dominates (253 of 291). Named repository sources (Jira, GitHub, GitLab, Jenkins, IDE) did not surface as majority labels: 183 studies have no event log, and the remaining 108 were coded `other/semi` rather than a named ALM tool. Process-mining algorithms are likewise rarely named beyond a generic "other" (191); stochastic methods are mostly absent or unspecified, with SPN (9) and DTMC (5) as the only named clusters above three studies. Forecast targets, when present, are seldom the SDLC operational metrics (lead/cycle/remaining time); `outcome` appears 17 times. Research type (Wieringa et al. [22]) is concentrated on solution proposals (265), with 18 evaluations and 8 philosophical papers. The remaining 1 417 corpus studies await the same coding; venues and publication types are reported separately once that pass (or a stratified sample) is complete.

4.4. RQ4: Integration and gaps

Table 10 gives the integration levels. Most studies (92.9%) use at most one technique family; 120 combine two; 2 combine three; none reaches L3.

Table 9: [PENDENTE] SDLC phases addressed, by technique family (classified studies). A study may address several phases.

SDLC phase	TF-PM	TF-ST	TF-FC	TF-MSR
Requirements and design	–	–	–	–
Development (commits, pull requests)	–	–	–	–
Quality (testing, review, defects)	–	–	–	–
Delivery (CI/CD, release)	–	–	–	–
Operations (incidents, maintenance)	–	–	–	–

Table 10: Integration levels of the 1710 studies.

Level	Definition	Studies	%
L0	At most one technique family	1589	92.9
L1	Two technique families (TF-PM+TF-ST 12; TF-PM+TF-FC 8; TF-ST+TF-FC 99)	119	7.0
L2	Three families; stages evaluated separately	2	0.1
L3	Three families coupled through specified interfaces, evaluated end-to-end	0	0.0
Total		1710	100.0

Process mining with stochastic modelling. 14 studies combine TF-PM and TF-ST, most of them published since 2014. They couple discovered process structure with stochastic reasoning for a single analytical concern. Incerto et al. show that variable-length Markov chains outperform eighteen stochastic conformance-checking techniques on ten of eleven benchmark logs [40]. López-Pintado and Dumas replace deterministic resource calendars with probabilistic ones discovered from event logs, reproducing empirical cycle-time distributions more closely [41]. Jalote and Kamma model programmer task sequences from IDE logs as first-order Markov chains [42]. None of these studies uses the stochastic layer to forecast a process metric.

Process mining with forecasting. 10 studies combine TF-PM and TF-FC (8 as a two-family combination, plus the L2 studies), all published since 2015. They approach prediction through process enhancement rather than an explicit stochastic model. Gupta combines inductive-miner discovery with predictive analytics to identify recurring failure patterns in maintenance workflows [43]. SIMPT extracts time-annotated process trees for interactive what-if simulation [44], a line extended with generative models [45]. Caldeira et al. derive process-complexity metrics from IDE logs and use them to predict product quality [46]. None reports a probability distribution over the completion time or remaining steps of an individual process instance.

The single three-family study. PRIMAD [47] builds a digital twin of the SDLC from Jira issue flows, combining pm4py-based discovery, an actor-based simulation core (Akka), and supervised learning (LightGBM). The PDF was read in full (Rodrigo, 2026-11-11). The authors state that they validated components independently and defer a full end-to-end test to a forthcoming article (Conclusions, p. 1684); formal stage interfaces are described only in prose. We therefore classify the study as L2, not L3. The component evidence is asymmetric: process mining is reported as descriptive profiling on a public Jira dataset (Results, p. 1684); the supervised model is

evaluated in production on 33 Scrum teams for *team routing* (accuracy 93%, macro-F_1 0.91), not for temporal process metrics and not on the public Jira log used in discovery; the Akka simulation is proposed and justified with a third-party performance benchmark, but Results report no simulation run of their own. Under IC4c as written (lead/cycle/remaining time, throughput, defect rates), the predictive target is borderline: next-state/resource or assignee classification rather than a duration or throughput forecast. We keep the TF-FC mark for consistency with predictive process monitoring in the corpus, and note the mismatch explicitly.

Gaps. Three gaps follow from Tables 6 and 10. First, the gap is one of integration, not of coverage: each family is well represented, but the PM and stochastic lines meet in 14 of 1710 studies. Second, forecasting is frequent (1341), but the PM–forecast link remains rare (10); stochastic models fed by aggregate statistics still account for 100 ST+FC studies, versus structure mined from event logs (8). Third, no study evaluates a multi-stage pipeline end-to-end on SDLC data.

5. Discussion

5.1. Two communities, one data source

The clearest pattern in the corpus is the succession of two communities over the same kind of data. Stochastic modelling of software processes, rooted in reliability engineering and process simulation, dominates the first two decades. Process mining, rooted in business process management, dominates the last one and increasingly uses repository data. Their methods are complementary: PM yields an empirically grounded structure, stochastic models yield distributions over paths and durations. Yet the corpus contains few studies that use one to feed the other. The PM-and-stochastic studies we found come mostly from the business-process side and are evaluated on generic benchmark logs [40, 41], a line that the stochastic process mining literature continues [48].

5.2. Research agenda

RA1: Standardised event-log construction for the SDLC. Frameworks that turn heterogeneous repository data into comparable event logs with cross-tool case correlation. In the corpus, Markov models of repository activity define their states manually and differently from study to study [24, 25, 26]. In the priority set, 183 of 291 studies have no event log (`NA`), and the 108 that do were coded `other/semi` rather than a named ALM source (Table 8).

RA2: Process-aware stochastic forecasting. Deriving the state space and transition probabilities of a stochastic forecasting model from a discovered process model, instead of from aggregate statistics. This is the least explored combination (TF-PM+TF-ST: 10 studies, none forecasting a process metric).

RA3: Process-derived features for learned predictors. Systematic use of conformance, rework, and path-entropy features in software engineering predictors, a direction suggested by [46, 43]. Forecast targets in the priority set are mostly `NA` (181) or generic `other` (93); operational SDLC targets (lead/cycle/remaining time) did not emerge as named categories in this pass.

Table 11: SPMF Analytical Depth × Technique Integration ($n = 1710$).

AD \ TI	TI-L0	TI-L1	TI-L2	TI-L3	Total
AD1	186	9	0	0	195
AD2	170	3	0	0	173
AD3	1201	106	1	0	1308
AD4	32	1	1	0	34
Total	1589	119	2	0	1710

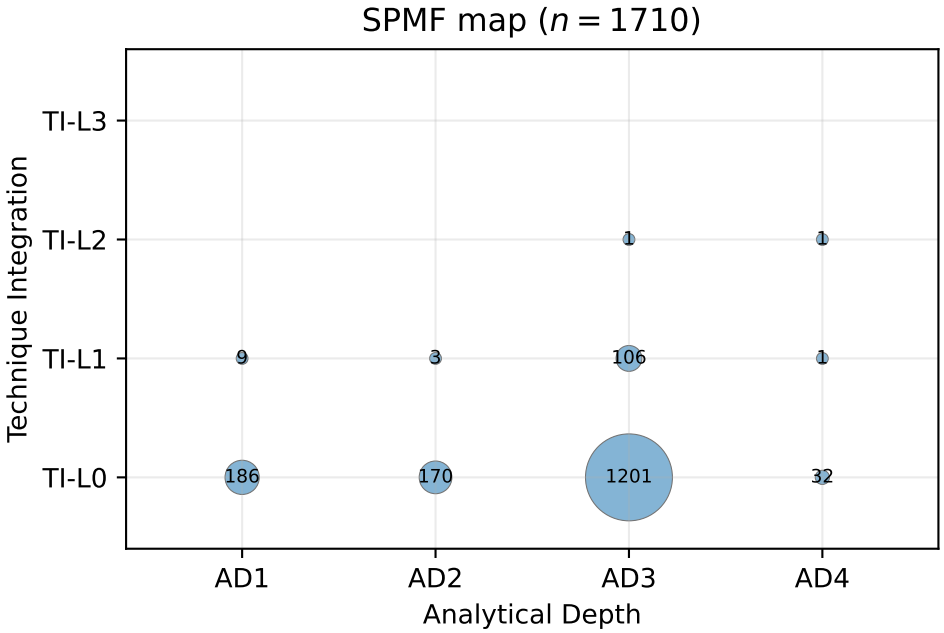


Figure 4: SPMF map: bubble area proportional to study count (AD × TI; $n = 1710$).

RA4: End-to-end benchmarks. Shared open-source benchmarks with temporal hold-out, so that multi-stage pipelines can be compared as a whole rather than per component (L2 → L3).

The Software Process Mining and Forecasting (SPMF) taxonomy positions studies on three axes: Analytical Depth (AD1 descriptive – AD4 distributional forecasting), SDLC Coverage (SC-R/D/Q/C/O), and Technique Integration (TI-L0–L3, identical to the L0–L3 scale of Table 10). We assigned all 1710 studies mechanically: TI from the family codes; SC from the RQ3 phase field when available, otherwise from title/family cues; AD from technique families, forecast targets, and title cues (PRIMAD was set to AD4 after full-text reading).

Most studies sit at AD3/TI-L0 (1201; predictive/estimation methods without a second technique family). AD1 (195) concentrates process-discovery work; AD2 (173) process simulation and diagnostic analyses; AD4 remains rare (34). On SDLC coverage (multi-label), development dominates (SC-D 1502), followed by quality (SC-Q 224); requirements, delivery/CI, and operations are sparse (SC-R 24, SC-C 27, SC-O 30). No study reaches TI-L3. The empty upper-right of Figure 4 (AD4×TI-L2/L3) is exactly the gap that RA2–RA4 target.

5.3. Implications for practitioners

The techniques in the corpus work on logs that already exist in issue trackers and version control, so the entry cost lies in event-log construction rather than in data collection. No study reports an end-to-end pipeline evaluated as one system. A practitioner who chains PM, a stochastic model, and a learned corrector today is combining individually tested parts whose joint behaviour is unknown.

6. Threats to Validity

6.1. Search

The strings may miss studies that use non-canonical terminology. The complementary MSR string, the frontier string, and the citation search mitigate this. The control check is weak evidence of completeness: the ten control studies were known when the strings were written, and three Scopus strings include the titles of control studies verbatim. The strings contain no block for learned forecasting on repository data, so studies that predict process metrics with machine learning without PM or stochastic terms are under-represented, and the TF-FC share is likely an underestimate. The review is limited to English, peer-reviewed publications; grey literature, theses, and preprints were not searched. IEEE Xplore exports were run in manual batches whose source string was not recorded.

6.2. Selection

The first pass used a language model. The model's Recall on the human-checked sample (0.630) was below the predefined criterion of 0.90. Much of the shortfall comes from the information available to it rather than from the model itself: for 72.4% of the primary-batch records the first decision was made on the title alone, because the Scopus export lacked abstracts, and the second pass read the PDF for 0.5% of the records. The model ran at the API's default temperature, so its decisions are not deterministic. For these reasons the first pass was used only to order the work, and every record it did not include was re-screened by humans. The residual threat lies in that human re-screen: agreement between the reviewers was moderate on the sample of the second batch ($\kappa = 0.494$), and the reviewers saw suggested labels while screening, which can anchor their decisions.

6.3. Classification

Family codes for the original includes were proposed by the language model and reviewed by the authors for every two-family study. A stratified sample of 55 L0 (single-technique-family) studies was re-coded by one author with 0% disagreement against the LLM codes on that sample; residual under-classification outside the sample cannot be ruled out. The L2/L3 judgement rests on the reading of one study (PRIMAD).

6.4. Construct

IC4b and IC4c admit Markov usage models for statistical testing and reliability-growth models of defect rates. These model the product rather than the development process and probably account for many of the 89 TF-ST+TF-FC studies of 1997–2013. The authors are developing an integrated forecasting pipeline; to limit the influence of this interest, the integration levels are defined by generic criteria, and the language-model prompt's reference to the authors' project name is disclosed in the replication package.

7. Conclusion

We mapped 1710 studies that apply process mining, stochastic modelling, or forecasting to software development data. The field has been shaped by two communities in succession: stochastic modelling until 2013 and process mining since 2014; they still meet in only 14 studies. After the Phase-3 ML block and snowballing to saturation, forecasting is common (78.2%), largely as FC-only work on repository data, while the PM–forecast link remains rare. Two studies carry all three technique families (PRIMAD [47] is the one we read in full for L2/L3); none couples them through specified interfaces evaluated end-to-end. The gap is therefore one of integration rather than of available techniques. Process-aware stochastic forecasting and end-to-end benchmarks are the two directions with the clearest support in the corpus.

The corpus, the classification, the prompts, and the screening records are available in the replication package (Zenodo, DOI 10.5281/zenodo.21939600).

CRediT authorship contribution statement

Rodrigo Almeida de Oliveira: Conceptualization, Methodology, Software, Investigation, Data curation, Formal analysis, Visualization, Writing – original draft, Writing – review & editing. **Juliano de Paulo Ribeiro:** Methodology, Validation, Investigation, Writing – review & editing. **Edson Emilio Scalabrin:** Conceptualization, Methodology, Supervision, Writing – review & editing, Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

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Declaration of generative AI in the writing process

During the preparation of this work the authors used Claude (Anthropic) to restructure and edit the manuscript and to check its arithmetic. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication. Separately, a language model was used in the conduct of the study as a first-pass screening aid (Section 2.4).

Data availability

The replication package (Zenodo, DOI 10.5281/zenodo.21939600) contains the search strings, de-duplication and enrichment scripts, screening prompts and schemas, per-record decisions at every step, the human verification and re-screen sheets, the list of included studies with their classification, and the completed SEGRESS checklist. The replication package ships `studies.json` (canonical classification of the $n = 1710$ included studies) and `figs.py`; any CSV export of the included list is derived from `studies.json` and is no longer versioned as `included_studies_506.csv`.

Appendix A. Search term lists

Principal string (population ∧ intervention).

```
("software engineering" OR "software development" OR "software process"
 OR "software lifecycle" OR "software life cycle" OR "SDLC"
 OR "development lifecycle" OR "development life cycle" OR "DevOps"
 OR "CI/CD" OR "continuous integration" OR "continuous delivery"
 OR "agile development" OR "agile process" OR "issue tracking"
 OR "bug tracking" OR "version control" OR "code review" OR "pull request"
 OR "software project" OR "software repository" OR "software maintenance"
 OR "software evolution")
AND
("process mining" OR "process discovery" OR "conformance checking"
 OR "workflow mining" OR "predictive process monitoring"
 OR "process prediction" OR "remaining time prediction"
 OR "process forecasting" OR "event log" OR "process simulation"
 OR "Monte Carlo simulation" OR "Markov chain" OR "Markov model"
 OR "stochastic process model" OR "transition probability"
 OR "absorbing chain" OR "lead time prediction" OR "cycle time prediction")
```

Complementary string (MSR → process modelling).

```
("mining software repositories" OR "software repository mining"
 OR "GitHub mining" OR "Jira mining")
AND ("process model" OR "Markov" OR "Monte Carlo" OR "stochastic"
 OR "lead time" OR "cycle time" OR "throughput" OR "transition matrix")
```

Frontier string (stochastic ∧ PM ∧ SE).

```
("Monte Carlo" OR "Markov chain" OR "stochastic simulation"
 OR "probabilistic forecast" OR "transition matrix")
AND ("process mining" OR "event log" OR "workflow mining"
 OR "process model" OR "business process")
AND ("software development" OR "software engineering" OR "software process"
 OR "DevOps" OR "SDLC" OR "issue tracker" OR "commit log"
 OR "build pipeline")
```

Appendix B. Included studies

Table B.12: The 1710 included studies (identifier in the replication package, publication year, technique and data families). PM = TF-PM, ST = TF-ST, FC = TF-FC, MSR = TF-MSR; [a] = added after the base corpus (human re-screen, Phase-3 ML block, snowballing, or F2.3 flips).

ID	Year	Families	ID	Year	Families	ID	Year	Families
23068d2c	1994	ST[a]	c7a601d8	1994	ST[a]	520023e9	1995	FC[a]
87bce30b	1995	PM+ST	2225ebb5	1996	FC[a]	6d68591e	1996	PM
eec839f7	1996	FC[a]	f9034283	1996	FC[a]	ff339f4b	1996	FC[a]
421b3545	1997	FC[a]	50144afd	1997	ST+FC	6115d05e	1997	ST[a]
8dea6481	1997	ST[a]	be7c6454	1997	PM+ST[a]	07e55287	1998	PM+ST
12f7f992	1998	ST+FC[a]	3f8fd941	1998	ST+FC	7f0d4fb9	1998	FC[a]
85002ffb	1998	ST[a]	17384cba	1999	FC[a]	20a00ebf	1999	FC[a]
28685daa	1999	FC[a]	66d00b19	1999	ST+FC	afd1b7aa	1999	ST+FC
b24b55e3	1999	ST[a]	d044ef55	1999	ST	eb05417e	1999	ST
eb526806	1999	PM	f453a92f	1999	ST	072ba70b	2000	FC[a]
46287754	2000	ST+FC	9c255853	2000	ST	a4f13cb1	2000	ST+FC
ab16b112	2000	ST	e82e259b	2000	ST[a]	f06ebc31	2000	ST
19b34231	2001	FC[a]	39e47477	2001	ST+FC	4155a4da	2001	ST
49233222	2001	FC[a]	50dda3a0	2001	ST+FC	5e48a99f	2001	FC[a]
72329237	2001	ST[a]	839d520d	2001	ST+FC	8ae72547	2001	ST+FC
b48dde83	2001	ST	da2f44fa	2001	ST+FC	e24af4e7	2001	ST[a]
e516f339	2001	FC[a]	f33d3c89	2001	ST	0743f004	2002	FC[a]
0b759736	2002	ST	1703f629	2002	FC[a]	181f6213	2002	FC[a]
363027c4	2002	ST[a]	743dcc43	2002	PM+ST[a]	7f8d2b91	2002	PM[a]
86be15b0	2002	FC[a]	883e1702	2002	ST+FC	92f75feb	2002	FC[a]
a12f49f1	2002	FC[a]	ad4f84a7	2002	ST+MSR	b1aabe94	2002	ST+FC
b75d202e	2002	ST+FC	c3e17c90	2002	FC[a]	dcf53abb	2002	ST[a]
e3a9143c	2002	ST[a]	e620e55e	2002	ST[a]	f9c9914d	2002	FC[a]
0cb1d54a	2003	FC[a]	1c928aae	2003	PM[a]	3487bbf5	2003	ST[a]
368dc22f	2003	ST[a]	39157e2c	2003	ST	3d0b754c	2003	FC[a]
46598602	2003	FC[a]	4b4df702	2003	FC[a]	4d84ffd3	2003	FC[a]
4deba73b	2003	FC[a]	6b16f9de	2003	ST[a]	6e4114dc	2003	ST+FC
78969f65	2003	ST	c35d9539	2003	FC[a]	cda3f627	2003	ST[a]
e9686368	2003	ST[a]	ebc6dbac	2003	FC[a]	1a7dfc0f	2004	FC[a]
25b10497	2004	FC[a]	3c810db6	2004	ST	46540576	2004	ST[a]
58996946	2004	ST	5ba78162	2004	FC[a]	6408e693	2004	PM+ST[a]
64c03c8d	2004	ST+FC	7f2dd217	2004	FC[a]	9e8d5d7d	2004	FC[a]
a05aee89	2004	FC[a]	ab99a53e	2004	PM+MSR	b37c10c1	2004	ST+FC
c03603a2	2004	FC[a]	c9512bbb	2004	ST+FC	fd7ca261	2004	ST[a]
0232276d	2005	ST[a]	11258fed	2005	FC[a]	190c11c4	2005	ST[a]
2fbf6d36	2005	ST+FC	33361b00	2005	ST[a]	3cc5a1cb	2005	ST
41a0e151	2005	FC[a]	48275fa8	2005	FC[a]	486ca310	2005	FC[a]
4a2d3698	2005	FC[a]	557b4102	2005	ST	5b51a065	2005	ST
6d37f6fe	2005	FC[a]	6d83b874	2005	ST+FC[a]	74a9a45a	2005	FC[a]
7bebca2e	2005	FC[a]	81929ce8	2005	FC[a]	88c18a84	2005	ST[a]
a20e7c58	2005	ST+FC	a27f51ca	2005	FC[a]	c94fc209	2005	ST+FC
d67f1905	2005	FC[a]	f330dbc0	2005	FC[a]	ff1faf6c	2005	FC+MSR[a]
063967eb	2006	ST+FC[a]	117db759	2006	ST[a]	24059629	2006	ST[a]
2c280053	2006	PM+MSR	307385ce	2006	FC[a]	35cf3bbf	2006	FC[a]
3c2136fd	2006	FC[a]	416c979c	2006	ST+FC	48e69187	2006	PM+MSR
53434e77	2006	ST+MSR	66c95d97	2006	ST[a]	6795e33c	2006	ST[a]
72bbaa1b	2006	FC[a]	74accbf5	2006	FC[a]	7766e7b4	2006	ST
78f76ab8	2006	FC[a]	836b5ef0	2006	ST+FC	8888cfdc	2006	ST
8e0eec4d	2006	ST+FC	9eff3fd0	2006	FC[a]	9f8b9dba	2006	ST+MSR
a9dd0444	2006	FC[a]	abc76a20	2006	FC[a]	ae4f4926	2006	ST
b5b1402b	2006	ST+FC	b5b6fcff	2006	FC[a]	b78523e4	2006	PM[a]
bd8b607e	2006	FC[a]	be942093	2006	PM	c3683c28	2006	PM+MSR
c89287f2	2006	FC[a]	cb6e46b8	2006	FC[a]	cdf1f104	2006	PM+MSR
d50fcf26	2006	ST	dbcd2435	2006	ST+FC	dc9483c1	2006	FC[a]
df3ad75a	2006	PM+MSR	ec94ba4d	2006	ST+FC	fe75ced8	2006	ST[a]
0078a735	2007	FC[a]	0c5b4491	2007	FC[a]	0e12a280	2007	FC[a]
16a201ed	2007	ST[a]	171085b9	2007	ST[a]	19f6b59f	2007	FC[a]
228961c5	2007	FC[a]	261f9099	2007	FC[a]	30769647	2007	FC[a]
3626fef4	2007	ST[a]	3ddbfd94	2007	PM	3ef9d4b3	2007	FC[a]
57195f58	2007	FC[a]	632586c5	2007	FC[a]	65bbe771	2007	FC[a]
747fc068	2007	ST+FC	8e41b716	2007	PM[a]	9870248b	2007	ST
a2bc654c	2007	FC[a]	a2d3ea2d	2007	FC[a]	a401497c	2007	ST[a]
a4ad0347	2007	ST[a]	a887b16d	2007	FC[a]	aa0da064	2007	ST+FC[a]
ac05d867	2007	ST	d1f084f4	2007	FC[a]	d7b22f54	2007	ST
df9a3f24	2007	FC[a]	e07c6819	2007	MSR[a]	e9a40e7a	2007	ST
e9d3ec8a	2007	FC[a]	ee3f437e	2007	FC[a]	f2ba79cc	2007	FC[a]

Continued on next page

ID	Year	Families	ID	Year	Families	ID	Year	Families
faea5ac1	2007	FC[a]	fba07e74	2007	FC[a]	fc1caae7	2007	FC[a]
fdd3425a	2007	FC[a]	01c715f4	2008	FC[a]	070fae7f	2008	FC[a]
0fcd5ece	2008	ST	2a8b96f3	2008	FC[a]	2d31d28f	2008	FC[a]
2fb4f513	2008	ST	37f008b1	2008	FC[a]	38625f3b	2008	ST[a]
38fae842	2008	FC[a]	3ae2f9fd	2008	ST[a]	3c0dd1ad	2008	ST+FC
40bc718e	2008	ST	4f530975	2008	FC[a]	52ee5271	2008	FC[a]
58475e17	2008	FC[a]	5c43edc7	2008	ST+FC	6269e223	2008	FC[a]
6f5e6f89	2008	ST[a]	7e6b3f9f	2008	PM	861c4959	2008	FC[a]
863bae01	2008	ST+FC	89fd72e8	2008	FC[a]	91493886	2008	FC[a]
afb37589	2008	FC[a]	b8b834e2	2008	ST	ba328b1c	2008	ST[a]
cb68ba73	2008	FC[a]	d8fc6aba	2008	FC[a]	deae809b	2008	ST+MSR
dfb2398b	2008	ST[a]	e080b18d	2008	ST+FC	e2d3c67e	2008	FC[a]
e5fa7505	2008	ST[a]	e6075fca	2008	ST[a]	edd4c661	2008	FC[a]
fb4a2cb3	2008	ST+FC[a]	fbf7e330	2008	ST[a]	008caf63	2009	ST+FC
0906c3aa	2009	ST[a]	20795cca	2009	FC[a]	2551d106	2009	FC[a]
2d148576	2009	FC[a]	30ff5eac	2009	FC[a]	31a8cb54	2009	ST[a]
35d428b0	2009	FC[a]	3b4591cd	2009	ST+FC	3e8a5f46	2009	MSR[a]
47de5b9b	2009	FC[a]	52c2c79d	2009	FC[a]	5e847cf3	2009	FC[a]
67b85117	2009	ST+FC[a]	6a71c8cc	2009	FC[a]	70af8211	2009	ST+FC
8f1fb42b	2009	ST[a]	917f238d	2009	PM	989891c5	2009	ST[a]
a31937db	2009	FC[a]	a7ff3f61	2009	FC[a]	a98f6c54	2009	ST[a]
b2c57fd3	2009	ST+FC	b594fa65	2009	FC[a]	b80334f7	2009	FC[a]
bd372642	2009	FC[a]	beb83053	2009	ST+FC	c0f6c7bd	2009	FC[a]
cfea47e6	2009	FC[a]	d12bfa4c	2009	FC[a]	d43051f3	2009	FC[a]
ddeca0e6	2009	PM	df61a7a1	2009	FC[a]	e1e6101d	2009	FC[a]
e856dbe4	2009	PM+MSR	ee5ccaa2	2009	PM	eeed2f5d	2009	ST[a]
fc8f325d	2009	ST+FC	fcc830f6	2009	ST	fd5bab59	2009	FC[a]
012922af	2010	FC[a]	02e1bb05	2010	FC[a]	0ab0a53e	2010	FC[a]
0ef6a5b4	2010	ST+FC	15d941d7	2010	FC[a]	1cd3c3f0	2010	FC[a]
1d754c2e	2010	ST[a]	26582491	2010	ST	2bc341c0	2010	FC[a]
3476b5db	2010	ST+FC	36ada7be	2010	FC[a]	3e19e5b7	2010	FC[a]
4a85a963	2010	FC[a]	4f3ac568	2010	FC[a]	52756657	2010	FC[a]
5357188e	2010	FC[a]	53c06c4c	2010	FC[a]	559a576d	2010	FC[a]
5f66b602	2010	ST+FC[a]	62936834	2010	ST+FC	6354fc81	2010	ST+FC
6c5aae2a	2010	FC[a]	7692b4f3	2010	FC[a]	7f497573	2010	ST+FC
7f8fbfcd	2010	FC[a]	839eaa6b	2010	FC[a]	865b13e0	2010	FC[a]
86d2ae2d	2010	FC[a]	86f0c466	2010	ST[a]	937705ca	2010	FC[a]
939e52ff	2010	FC[a]	980c26fa	2010	FC[a]	a210fdd3	2010	ST[a]
a222ef2c	2010	PM+MSR	a5c1650b	2010	MSR[a]	a7082664	2010	FC[a]
a7e713c2	2010	FC[a]	a894367d	2010	FC[a]	afb8f80d	2010	FC[a]
b835ddc7	2010	FC[a]	ba2ff831	2010	ST	bafb58a0	2010	PM+MSR
c113bf1b	2010	PM+MSR	c72f823c	2010	ST	ca22bfe7	2010	ST[a]
cd860b9f	2010	ST[a]	d55dd861	2010	FC[a]	d96a3161	2010	PM+MSR
e93c382e	2010	FC[a]	f22983b3	2010	FC[a]	ffe69110	2010	FC[a]
0132f840	2011	FC[a]	04e0e983	2011	ST[a]	0b2f6ca9	2011	FC[a]
104187af	2011	FC[a]	155125ff	2011	FC[a]	16ec93d7	2011	ST[a]
18910ceb	2011	FC[a]	1caa9a9e	2011	FC[a]	1e4da643	2011	FC[a]
1edfbbfa	2011	FC[a]	2528f10d	2011	FC[a]	266210a7	2011	ST
27e504af	2011	FC[a]	296c473a	2011	FC[a]	2de8a9af	2011	FC
3155ca7a	2011	FC[a]	32aacfa7	2011	FC[a]	35bcb278	2011	ST+FC[a]
37582a83	2011	FC[a]	386325c8	2011	FC[a]	3d174d69	2011	FC[a]
3f5f23a9	2011	FC[a]	4316248d	2011	PM	460fa1fc	2011	FC[a]
47ed5bf0	2011	ST+MSR	4a072003	2011	FC[a]	4d088b6b	2011	ST[a]
4f2424db	2011	FC+MSR[a]	51b4ae2b	2011	ST+FC	576b8543	2011	FC[a]
57d2445c	2011	ST+FC	5c322411	2011	FC[a]	5c3f99ea	2011	ST[a]
5c86719e	2011	FC[a]	5e335d0b	2011	FC[a]	602d4cc6	2011	ST[a]
6dbfdf97	2011	FC[a]	6e574443	2011	FC[a]	704a0d34	2011	ST[a]
7206bd3c	2011	FC[a]	7a18f4ee	2011	FC[a]	7afcd1b2	2011	FC+MSR[a]
7eaf7eb3	2011	ST	7fd7fade	2011	FC[a]	84315802	2011	ST
917d9592	2011	ST	93f4daf3	2011	FC[a]	986ed4b1	2011	FC[a]
9ae84d72	2011	PM[a]	9f26ca4a	2011	FC[a]	a295c1b6	2011	FC[a]
aaa79bcd	2011	FC[a]	b61ea92e	2011	ST+FC	bc57ea5b	2011	FC[a]
c24db118	2011	PM	c4269b60	2011	FC+MSR[a]	c4703958	2011	FC[a]
c517c93f	2011	PM+MSR	c7682ec2	2011	PM+MSR	c815bcc6	2011	ST[a]
cafaba8c	2011	FC[a]	cccf4e66	2011	FC[a]	ce92cc23	2011	FC[a]
d098c91d	2011	PM	d556b5d1	2011	FC[a]	d7bc3e1a	2011	FC[a]
de24553f	2011	FC[a]	debdebc2	2011	ST[a]	e0cd71ab	2011	FC[a]
e39e4f39	2011	FC[a]	e8e53811	2011	FC[a]	ee504329	2011	FC[a]
efab3f7f	2011	ST[a]	f047f55e	2011	FC[a]	f450a0aa	2011	FC[a]
f516d8b6	2011	FC[a]	0202e682	2012	FC[a]	03fd9dfc	2012	FC[a]
076d92c4	2012	FC[a]	0c10d633	2012	ST+FC	0d306f3b	2012	FC[a]
0e15e613	2012	MSR[a]	0e708ae8	2012	MSR[a]	1099ff04	2012	FC[a]

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1303e41e	2012	ST[a]	1bfc6bec	2012	FC[a]	1f3777ef	2012	FC[a]
21121cbf	2012	ST+MSR[a]	214d17c4	2012	ST[a]	214efe16	2012	FC[a]
23ec55bd	2012	ST	2c7144d6	2012	FC[a]	2c7ec6c7	2012	FC[a]
2f7a6894	2012	FC+MSR[a]	326da20f	2012	ST	33cdea14	2012	ST[a]
37bba7f6	2012	FC[a]	3ee20edf	2012	FC+MSR[a]	3f09473c	2012	ST[a]
4634263a	2012	FC[a]	46cbe525	2012	FC[a]	489b63cb	2012	FC[a]
4bc22efd	2012	FC[a]	555f2c80	2012	FC[a]	5593f550	2012	FC[a]
5aa1ae73	2012	FC[a]	5ae29e8c	2012	FC[a]	5c7a03c0	2012	FC[a]
5ffe33d2	2012	FC[a]	66a32124	2012	ST+FC	6c89be51	2012	FC[a]
72fb141c	2012	FC+MSR[a]	7453dad6	2012	FC[a]	79eac7e5	2012	FC[a]
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a3c061ae	2012	ST+FC	a8a3e15d	2012	FC[a]	ab9983cb	2012	FC[a]
ae20e68e	2012	PM	ae2ff7cf	2012	FC[a]	aea9f86e	2012	FC[a]
aef9dbee	2012	FC[a]	b58bf16e	2012	FC[a]	b888412a	2012	FC[a]
b8d51dfd	2012	FC[a]	b94c32dd	2012	FC[a]	bbc13de4	2012	PM+MSR
bcce3152	2012	FC[a]	bfe1fbcc	2012	FC[a]	c98ee85a	2012	FC[a]
c9ee07ba	2012	FC[a]	ca00050f	2012	ST+MSR	d3689e89	2012	FC[a]
d4601a2f	2012	MSR	da28afcb	2012	ST[a]	dbc78b51	2012	FC[a]
df22b7bc	2012	FC[a]	dfe889ba	2012	FC[a]	e6eada40	2012	FC[a]
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fa9fe762	2012	FC[a]	fcbe1fac	2012	FC[a]	ff19d97c	2012	FC[a]
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09363023	2013	FC[a]	0d29ff1f	2013	FC[a]	14dbc2fa	2013	FC[a]
17a4707b	2013	FC[a]	18289398	2013	FC[a]	1f58d29f	2013	ST+FC[a]
203ec332	2013	FC[a]	264019f3	2013	FC[a]	2784f4ea	2013	FC[a]
2d93db80	2013	FC[a]	322484bf	2013	FC[a]	3b614f91	2013	ST+FC
3d6aeed8	2013	MSR[a]	40b67211	2013	FC[a]	43773183	2013	ST[a]
455b830c	2013	FC[a]	4ba174fc	2013	FC+MSR[a]	4c5b8652	2013	FC[a]
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5f9bb212	2013	FC[a]	60e4fba6	2013	MSR[a]	6789e54a	2013	FC+MSR[a]
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84407144	2013	FC[a]	86c11e29	2013	FC[a]	88b8156e	2013	FC[a]
8ea2246c	2013	FC[a]	9174f5b6	2013	FC[a]	95fbffdc	2013	FC[a]
97603c0b	2013	FC[a]	9d553716	2013	FC[a]	a257388d	2013	FC[a]
a8594e55	2013	FC[a]	a8b7ed5a	2013	FC[a]	aeae5ed0	2013	FC[a]
b4540e0e	2013	FC[a]	b66fac38	2013	FC[a]	b721525d	2013	FC[a]
bd0d0222	2013	FC[a]	c195758e	2013	FC[a]	cc3e46a9	2013	FC[a]
cd17b2ad	2013	ST[a]	cd32f232	2013	FC[a]	d2b57e47	2013	FC+MSR[a]
d5665995	2013	FC[a]	d892a8fb	2013	FC[a]	ddc8cc8c	2013	FC[a]
ecbed8c4	2013	FC[a]	edb05739	2013	FC[a]	edda2671	2013	FC[a]
faa981c4	2013	FC[a]	fdbc7853	2013	FC[a]	fed24e09	2013	FC[a]
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0c80db13	2014	FC[a]	1282204f	2014	FC[a]	1363926c	2014	ST+FC
149dedd4	2014	FC[a]	14bd5a7f	2014	FC[a]	183b6bb9	2014	ST+FC[a]
1de657c1	2014	PM+MSR	2099ce3a	2014	FC[a]	229f1656	2014	FC[a]
2365ec87	2014	FC[a]	25969842	2014	FC[a]	26b762ac	2014	FC[a]
27ca7b0a	2014	FC+MSR[a]	2a7de581	2014	ST+FC	2f1e25da	2014	ST[a]
3231578f	2014	PM	358ea1c9	2014	FC[a]	3726ebce	2014	FC[a]
3bd2d170	2014	FC[a]	3e13bdae	2014	FC[a]	42e8f821	2014	FC[a]
4748abd8	2014	FC[a]	477393d4	2014	FC[a]	49527f55	2014	FC+MSR[a]
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55def679	2014	FC[a]	566e24e8	2014	FC[a]	586b96ab	2014	FC[a]
5a466249	2014	FC[a]	5a602862	2014	FC+MSR[a]	5d027dd1	2014	FC[a]
5faae738	2014	FC[a]	5fc630db	2014	FC[a]	5fe2fe46	2014	ST[a]
68fb9a94	2014	PM[a]	6927a5a3	2014	FC[a]	6952fcb7	2014	FC[a]
6b273691	2014	FC[a]	6f697869	2014	FC[a]	6fd30d7e	2014	PM+MSR
70c7203e	2014	FC[a]	76e991ee	2014	FC+MSR[a]	7ac425f7	2014	FC[a]
7f7591d0	2014	FC[a]	7fdeb0fa	2014	FC[a]	829f8c13	2014	PM+MSR
88ee1d90	2014	ST+FC[a]	8c42d0fe	2014	PM[a]	8c928c6a	2014	FC+MSR[a]
8cb831d9	2014	FC[a]	90ce9c17	2014	FC[a]	932f2a18	2014	PM
98c7bb26	2014	FC[a]	99310b4c	2014	FC[a]	9dd53fdd	2014	FC[a]
a13cfb0c	2014	FC[a]	a16b863d	2014	FC[a]	ad250ae5	2014	PM
ad5c0b40	2014	FC+MSR[a]	b117b377	2014	FC[a]	b5a40ee4	2014	FC[a]
b9b3d835	2014	FC[a]	b9d7b3e8	2014	FC[a]	bc871cea	2014	FC[a]
bcdd7b6f	2014	FC[a]	c1575581	2014	FC[a]	c2dae430	2014	FC[a]
c50ffd51	2014	FC[a]	c6366ed4	2014	FC[a]	c854882d	2014	FC[a]
c8e54484	2014	FC[a]	c8ea828e	2014	PM	c9cdb64d	2014	PM
ce72fe95	2014	FC[a]	cef0d2f7	2014	FC[a]	d1ed6154	2014	PM+MSR
d6b59a61	2014	FC[a]	d6d61646	2014	FC[a]	d82592db	2014	FC[a]
db3af2f6	2014	FC[a]	defccca9	2014	FC[a]	e032af1f	2014	FC[a]
e2533ddb	2014	FC[a]	e36fe110	2014	FC[a]	e58074e0	2014	PM

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f1fc3f03	2014	FC[a]	f27bbb48	2014	FC[a]	f28af826	2014	FC[a]
f3b5dc7a	2014	FC[a]	f6edde4a	2014	ST	fe09a77e	2014	FC[a]
fe225f50	2014	FC+MSR[a]	02f7a03b	2015	FC[a]	02f96e2f	2015	FC[a]
058b73dc	2015	FC[a]	05b677ed	2015	FC[a]	087be32f	2015	FC[a]
0d56225b	2015	FC[a]	12fbf63a	2015	ST[a]	164bff33	2015	FC[a]
18ed6f12	2015	FC[a]	1f99d374	2015	FC[a]	261f6383	2015	ST+FC
2647ea07	2015	PM	27a05621	2015	FC[a]	2b497476	2015	FC[a]
2ee3dc10	2015	PM+MSR	32027321	2015	FC[a]	328123bc	2015	PM+ST+FC[a]
329c586c	2015	FC[a]	35f7dbe4	2015	FC[a]	3a39fa54	2015	FC[a]
3ac74336	2015	ST+FC	3ad60828	2015	FC[a]	3ad9cd69	2015	ST+FC
3d379233	2015	PM+MSR	405b5b88	2015	FC[a]	4b2c7904	2015	FC+MSR[a]
4eeb076b	2015	PM+FC[a]	508f3b5b	2015	PM	5202932a	2015	PM[a]
582a391e	2015	FC[a]	5af6706b	2015	PM[a]	5d68fa29	2015	FC[a]
5eba4c4b	2015	FC[a]	605e3a34	2015	FC+MSR[a]	6069e7b2	2015	FC[a]
769bbb49	2015	FC+MSR[a]	786d57aa	2015	ST[a]	789cf287	2015	FC[a]
7a242c5c	2015	FC[a]	80efe394	2015	FC+MSR[a]	82b3401d	2015	PM
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98875adb	2015	FC+MSR[a]	9c546e13	2015	FC[a]	9cf74956	2015	FC[a]
9e0287a2	2015	FC[a]	a4d8ff91	2015	FC[a]	a7346c1d	2015	FC[a]
a83fbf3b	2015	FC[a]	aa3ce899	2015	PM+MSR	af77026f	2015	ST[a]
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c5aecde4	2015	FC[a]	c64d551d	2015	FC[a]	c7ed395a	2015	ST
d2e1b306	2015	PM+FC+MSR	d3cdd485	2015	FC+MSR[a]	d87c2315	2015	FC[a]
db0205c3	2015	FC[a]	de7755a8	2015	ST[a]	e210b995	2015	FC[a]
e23bbc10	2015	PM	ee562777	2015	PM+MSR	ee7b6d6f	2015	FC[a]
eea29299	2015	PM	efbaa4df	2015	FC+MSR	f2c78b19	2015	ST[a]
f356af5c	2015	ST+FC	f822c54d	2015	ST[a]	fa561e12	2015	PM+MSR
00ce8cfe	2016	PM	08c8604c	2016	FC[a]	16e31373	2016	PM
1c22f973	2016	FC[a]	1cf54e25	2016	FC+MSR[a]	1e3aae6f	2016	FC[a]
22670b24	2016	FC[a]	263eb268	2016	FC[a]	2b343650	2016	PM+MSR
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375de3a2	2016	FC[a]	37b22a16	2016	PM	3e0fcce0	2016	FC[a]
3e32c794	2016	ST+FC[a]	4541f643	2016	PM	4dc479a6	2016	PM
4f816525	2016	PM	5867eec5	2016	FC+MSR[a]	58d41ee9	2016	FC[a]
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6aa663f2	2016	FC[a]	6acde3bb	2016	FC[a]	6c6e27e2	2016	FC[a]
6e93d023	2016	FC+MSR[a]	6ecc75bd	2016	FC[a]	6f0a72fc	2016	FC[a]
74b26227	2016	FC[a]	76c610d5	2016	FC[a]	7ab40121	2016	ST+FC
7ff6cce0	2016	ST[a]	81e495b1	2016	FC[a]	84dc0c7f	2016	FC[a]
855ab5cc	2016	FC[a]	867a8b8f	2016	PM+MSR[a]	8740e969	2016	FC[a]
9334b926	2016	FC[a]	971ee8d0	2016	FC[a]	98d6af64	2016	FC[a]
9e6710b1	2016	FC[a]	a334d7c8	2016	FC[a]	a94c0eb9	2016	FC[a]
ad1c626a	2016	FC[a]	aeed4e9a	2016	FC[a]	af39dae4	2016	FC[a]
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b269731f	2016	ST[a]	b7fd840a	2016	ST	b9a91686	2016	FC[a]
bce026ce	2016	PM+MSR[a]	c2612028	2016	FC[a]	c42be0aa	2016	PM+ST
c77dd795	2016	FC+MSR[a]	c91d8906	2016	FC[a]	cbb476ba	2016	FC[a]
cd045558	2016	FC+MSR[a]	d1a8ea2e	2016	FC[a]	d33131d0	2016	FC[a]
d52477eb	2016	FC[a]	d78d3ca7	2016	FC[a]	d7953e93	2016	FC+MSR[a]
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e314074a	2016	FC[a]	e389b141	2016	FC[a]	ebf27b98	2016	FC[a]
ed81f924	2016	ST+FC[a]	eefc7305	2016	PM+ST	ef286e55	2016	FC[a]
f29c1e7d	2016	FC[a]	01f8cab1	2017	ST[a]	08d64489	2017	PM
0d31db29	2017	PM	0f01bc02	2017	FC[a]	112983ec	2017	FC[a]
1145c10d	2017	FC	133dec31	2017	ST[a]	160fface	2017	PM+MSR
16b51f01	2017	FC[a]	1d2d9238	2017	PM[a]	20c17d18	2017	ST+FC
25217fd5	2017	FC+MSR[a]	258b3c55	2017	FC[a]	2a92877a	2017	FC+MSR[a]
2acf06bd	2017	FC[a]	2c2d9e02	2017	FC+MSR[a]	2cfeaa56	2017	FC[a]
2dcc2981	2017	FC+MSR[a]	2e76e921	2017	PM	2f707065	2017	FC[a]
3a4c232a	2017	FC[a]	3d967450	2017	ST+FC	3ef399e6	2017	PM
41df3b08	2017	FC[a]	438b86a0	2017	FC[a]	4c73b15f	2017	FC
4ce9376c	2017	FC[a]	4ee1a8dd	2017	FC[a]	4fd2327e	2017	ST
500dc387	2017	FC[a]	52ecbd82	2017	FC+MSR[a]	5694f6fa	2017	FC[a]
58d1fb60	2017	FC[a]	59674b80	2017	FC+MSR[a]	5b014261	2017	FC+MSR[a]
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6213e062	2017	PM[a]	6217173e	2017	MSR[a]	63bd2d51	2017	FC[a]
64a5aad2	2017	FC[a]	64df6884	2017	FC[a]	66bf44d0	2017	FC[a]
691474c5	2017	FC[a]	6be6f20c	2017	ST+MSR[a]	6f007eeb	2017	ST+FC[a]
71b0b968	2017	FC[a]	74a0f623	2017	FC+MSR[a]	7a0e1214	2017	FC[a]

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856ea57a	2017	FC[a]	9061a5a2	2017	FC+MSR[a]	90e4a822	2017	FC[a]
92e46adb	2017	FC[a]	9e609be4	2017	FC[a]	9ed217f3	2017	FC[a]
9fcb4932	2017	FC[a]	a21c9dff	2017	ST[a]	a63597c8	2017	FC+MSR[a]
b2727486	2017	FC[a]	b58a2be8	2017	FC[a]	b5ea4fdd	2017	ST
b940a63a	2017	PM	bc54c3b6	2017	ST+FC	bd831ab6	2017	FC[a]
bf038ef2	2017	FC[a]	c51bae0c	2017	FC[a]	c58f7e3f	2017	FC[a]
c5bdca77	2017	FC+MSR[a]	c612882b	2017	FC+MSR[a]	c63395c8	2017	FC[a]
c65a4709	2017	FC[a]	c695c51f	2017	PM[a]	c964bc1a	2017	FC+MSR[a]
ca6989c0	2017	PM	ca6ae0ac	2017	FC[a]	d0e040e1	2017	PM+MSR
d1d1bc53	2017	FC[a]	d1e0f7ed	2017	FC[a]	d4a1d75c	2017	MSR
db0cbc8e	2017	FC[a]	dcbf27d8	2017	PM+FC+MSR	de02c8c3	2017	FC[a]
ded31b9b	2017	FC[a]	e24b6268	2017	FC[a]	e31c5676	2017	FC[a]
e9351089	2017	FC[a]	ec38f0ba	2017	PM+MSR	f0cb7ba4	2017	FC[a]
f5a86cc5	2017	FC+MSR[a]	f6e8c10c	2017	FC[a]	f858387d	2017	FC+MSR[a]
fb4ec2ea	2017	FC[a]	fbc469cf	2017	FC+MSR	fc3f666d	2017	FC[a]
febd37cc	2017	FC[a]	034cab3c	2018	FC[a]	04ab260e	2018	PM
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0b91f848	2018	FC[a]	0d4ebe86	2018	PM+MSR[a]	14ddbda5	2018	FC[a]
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23b0e415	2018	FC[a]	2ab2d233	2018	PM[a]	2ec898ed	2018	FC[a]
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3eaa4fbe	2018	FC[a]	4082a5b1	2018	PM	40e766d1	2018	FC[a]
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579522da	2018	FC[a]	57f648b0	2018	FC[a]	590acd92	2018	FC[a]
5c4c0e7f	2018	PM+FC+MSR	5d16be8a	2018	FC+MSR[a]	5d311a9e	2018	FC[a]
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65e5d58c	2018	ST[a]	65f2abf7	2018	FC[a]	660247e6	2018	FC+MSR[a]
67efc6a5	2018	FC+MSR[a]	6f9a5a08	2018	PM	704e89aa	2018	FC[a]
7129d07c	2018	FC[a]	774038d1	2018	FC[a]	7a78ddb4	2018	FC[a]
7cc61067	2018	ST[a]	800ef1a0	2018	FC[a]	852e6787	2018	FC[a]
8962430e	2018	FC+MSR[a]	8dac361c	2018	FC[a]	9337fa05	2018	FC[a]
9629d7e4	2018	FC[a]	991a4fc8	2018	FC[a]	9a40c123	2018	FC[a]
9b19cae9	2018	FC[a]	9e341b7f	2018	ST+FC	a090e0ab	2018	FC[a]
a20d9dec	2018	FC[a]	aa5ff1ea	2018	FC[a]	ab941422	2018	FC[a]
ac960d23	2018	FC[a]	aea4d2b9	2018	FC[a]	af251ed3	2018	FC[a]
b5349418	2018	FC[a]	b5d78dbb	2018	FC[a]	c415ab5d	2018	FC[a]
c54c5c70	2018	FC[a]	c56db11b	2018	FC[a]	ca6319b0	2018	FC[a]
ccca55c1	2018	FC[a]	d0c47271	2018	PM	d3493587	2018	FC+MSR[a]
dcc44897	2018	FC[a]	e12fff7c	2018	PM+MSR	e44d70d5	2018	FC+MSR[a]
edffd6bd	2018	FC[a]	efe72cc3	2018	FC[a]	fb5ff164	2018	FC[a]
005f0816	2019	FC[a]	0745f59f	2019	FC[a]	0dbf542b	2019	FC[a]
0f8da0ac	2019	FC[a]	1068b753	2019	ST+FC[a]	10b1fc22	2019	FC[a]
10d13363	2019	FC[a]	13473087	2019	FC+MSR[a]	16082053	2019	FC[a]
1659baf7	2019	FC[a]	18ce22f8	2019	FC[a]	1a642953	2019	FC[a]
1e7e7044	2019	FC+MSR[a]	1efe05a6	2019	FC[a]	283649c0	2019	FC[a]
28b6c6e9	2019	FC[a]	2cdc2aa0	2019	FC[a]	2d369fe9	2019	PM+ST
2e10e5d3	2019	FC[a]	32a1c395	2019	FC[a]	337b566a	2019	FC[a]
363f7789	2019	FC+MSR[a]	391cd760	2019	FC+MSR[a]	3bc2adaa	2019	FC[a]
422bb6d5	2019	FC[a]	440e98e2	2019	FC[a]	4ae03672	2019	FC[a]
4b92ed87	2019	FC[a]	4ba255cc	2019	FC+MSR[a]	537bbfd0	2019	ST+FC
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6863fe55	2019	FC+MSR[a]	6bb009be	2019	FC[a]	6cb59310	2019	PM
73ca720b	2019	FC+MSR[a]	73df5473	2019	ST	76b62e3c	2019	FC[a]
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9f6781fc	2019	FC[a]	a0211989	2019	FC[a]	a3a4c18c	2019	FC[a]
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c903b289	2019	FC[a]	d59e0082	2019	FC[a]	d5aff523	2019	PM+MSR
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f039e75e	2019	FC[a]	f16294f2	2019	PM+MSR	f302d687	2019	FC+MSR[a]

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0b72b67c	2020	FC+MSR^a	0cb156c2	2020	FC^a	1078bc2f	2020	FC^a
11ef9e59	2020	FC^a	13e29a4f	2020	FC^a	15c9808b	2020	PM
1d075126	2020	FC^a	2034a3b5	2020	FC^a	21974694	2020	FC^a
23754909	2020	FC^a	24564602	2020	FC^a	25b131a4	2020	FC+MSR^a
286822d6	2020	FC^a	2ac6a54f	2020	FC+MSR^a	2ce3aaae	2020	FC+MSR^a
2fd41ca7	2020	FC^a	340c0c24	2020	ST+FC	359d0c3e	2020	FC^a
463ce392	2020	FC^a	4683f278	2020	FC^a	4bf3a2b2	2020	FC+MSR^a
5029f905	2020	PM+MSR	544adfb9	2020	FC+MSR^a	580afeeb	2020	FC^a
5ba2549e	2020	FC+MSR^a	5c5bc65c	2020	FC^a	5eee7a36	2020	FC^a
649e8c83	2020	FC^a	65d6ff16	2020	FC^a	662f2b61	2020	FC+MSR^a
68f7fbe6	2020	FC+MSR^a	6bc47fa4	2020	FC+MSR^a	6e572cbb	2020	PM+MSR
6ff7b033	2020	FC^a	71e4ea19	2020	FC^a	76a781c6	2020	ST^a
7a71754a	2020	FC^a	7b642237	2020	FC+MSR^a	7cf00385	2020	PM+MSR
7e590666	2020	FC^a	8347d399	2020	FC+MSR^a	8a97d0cf	2020	ST+FC
8de6dcfa	2020	ST+FC	9793ac79	2020	FC^a	97d0c37f	2020	FC^a
99403a03	2020	FC+MSR^a	99736057	2020	PM+MSR	a172dd32	2020	FC^a
a2cf047c	2020	FC^a	a8cee3f0	2020	FC^a	aed9a2a9	2020	FC^a
b0bb5025	2020	FC+MSR^a	b115902f	2020	FC^a	b47d62ac	2020	FC^a
b7022ac2	2020	PM+MSR	b799fa10	2020	PM+MSR^a	bbc11d56	2020	ST^a
bc5255b3	2020	FC+MSR^a	bc63ae20	2020	FC+MSR^a	bde9904f	2020	FC^a
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ead7253c	2020	FC+MSR^a	f08136ac	2020	FC^a	f1233267	2020	FC^a
fe6aee24	2020	FC^a	fec7d66f	2020	FC^a	0999fe57	2021	ST^a
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0dd5604d	2021	FC+MSR^a	0f428f0c	2021	FC+MSR^a	15b3a419	2021	FC^a
16644fb0	2021	PM+MSR	16656db3	2021	FC^a	2188570a	2021	FC+MSR^a
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346e6a05	2021	FC+MSR^a	348377f4	2021	FC+MSR^a	374c6f47	2021	FC^a
38694b4e	2021	ST+FC	3919db26	2021	FC^a	3a0d3e8f	2021	FC^a
3aed577c	2021	FC^a	413f6dbf	2021	FC^a	449941a5	2021	FC^a
45e9a7c0	2021	FC+MSR^a	467b072b	2021	FC^a	483d7725	2021	ST+FC
4852a4fe	2021	FC^a	52a6ac1d	2021	FC^a	53d860d0	2021	FC^a
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73193c4a	2021	FC^a	76902719	2021	PM^a	77fa80cd	2021	PM
7b0b22ea	2021	PM+ST	812a4b8f	2021	FC^a	81a1d12d	2021	FC^a
8d2c0b03	2021	ST^a	9085dddf	2021	FC^a	90a81935	2021	FC+MSR^a
91e2f1e5	2021	FC^a	94868b58	2021	FC^a	977e32e3	2021	FC^a
9c4cc898	2021	PM+ST	9cccc3a3	2021	FC^a	9e450f55	2021	PM+ST
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a86b1042	2021	PM	ab4bbb63	2021	FC^a	abfd6b2e	2021	PM
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e767bde6	2021	FC^a	e89188a6	2021	ST+FC	e8b6d34c	2021	FC^a
ea1c9c15	2021	FC+MSR^a	eb2243dd	2021	FC^a	ebbe7cec	2021	FC^a
f06925e7	2021	FC^a	f27afe4e	2021	FC+MSR^a	f36a08fe	2021	ST+FC
f59b10d5	2021	FC	f7815b3d	2021	FC^a	fa1d225c	2021	FC+MSR^a
fb6f87a0	2021	FC^a	fc04d830	2021	ST^a	ffa8885e	2021	PM+MSR
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0587b264	2022	PM+MSR	0d0c0c27	2022	FC^a	0e197650	2022	FC^a
11658ee6	2022	FC^a	12343eec	2022	PM^a	168ed6b3	2022	FC^a
16c21c78	2022	FC^a	1866175f	2022	FC^a	1ad50ad1	2022	FC+MSR^a
1c6b409a	2022	FC+MSR^a	1e0b13ad	2022	FC^a	1fb6e8b4	2022	FC^a
1fd95ebc	2022	PM^a	21ddf169	2022	FC^a	22412f71	2022	PM+MSR^a
24d2c572	2022	FC^a	276f0a66	2022	FC^a	2845d106	2022	FC^a
2aa5e982	2022	FC^a	2bb1cf31	2022	FC^a	2cfc1a72	2022	FC^a
31da514c	2022	FC^a	34d52a2f	2022	FC^a	350880c5	2022	FC^a
3be229e7	2022	FC^a	3c502f7a	2022	FC^a	3cfa279d	2022	FC^a
41265418	2022	FC^a	43a78099	2022	FC^a	4769150f	2022	FC^a
47b0719c	2022	FC^a	4b34f7a5	2022	PM+FC	4e1875d9	2022	FC^a
5211dc1a	2022	FC+MSR^a	5949a370	2022	FC+MSR^a	5b3197ef	2022	FC^a

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711d2c92	2022	FC+MSR[a]	71b47b4b	2022	FC+MSR[a]	776170ae	2022	FC[a]
780addad	2022	MSR[a]	7a92806a	2022	ST+MSR[a]	7ad876c1	2022	PM[a]
7c4d9287	2022	FC[a]	7e76d2d6	2022	FC+MSR[a]	7eebcd00	2022	PM
83f25bf6	2022	FC+MSR[a]	88087b27	2022	PM	8c819f87	2022	FC+MSR[a]
8d5ed3d3	2022	FC[a]	92fa0290	2022	PM[a]	9338460f	2022	FC[a]
9370da82	2022	FC[a]	9434fa58	2022	FC[a]	a62b80dc	2022	FC[a]
a78db39c	2022	FC[a]	a82a1e30	2022	FC+MSR[a]	ae9983fa	2022	ST+FC
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b42eb4db	2022	FC[a]	b5ffb64d	2022	PM	b8fc130d	2022	PM+MSR
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ca9bdfc4	2022	FC[a]	cd664b7e	2022	FC+MSR[a]	cf61d53e	2022	FC[a]
d091ea64	2022	FC[a]	d8102d3e	2022	FC[a]	ddc43b07	2022	FC+MSR[a]
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1b145787	2023	FC[a]	1b7688b7	2023	PM	1efe6b8c	2023	PM+FC+MSR[a]
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26640417	2023	FC+MSR[a]	26709b6a	2023	FC+MSR[a]	267f242f	2023	FC[a]
295668be	2023	FC[a]	29edb55d	2023	FC[a]	2cc2226b	2023	FC[a]
311ad597	2023	FC[a]	35fe2c09	2023	FC[a]	3730ecc7	2023	FC[a]
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667f1705	2023	ST+MSR	6870a8ba	2023	FC[a]	6887dcec	2023	FC[a]
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34a0c931	2024	FC+MSR[a]	369baeae	2024	FC[a]	36ecb914	2024	FC+MSR[a]
36f95281	2024	FC[a]	373ad20e	2024	PM	37e6566e	2024	FC[a]
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6e1f98bc	2024	FC[a]	6e242e6b	2024	FC[a]	6ea37913	2024	FC[a]
6ff2b7f2	2024	ST+MSR	6ffcab24	2024	MSR[a]	7221e872	2024	ST[a]
752d548f	2024	FC[a]	756b8a78	2024	FC+MSR[a]	7642faee	2024	FC[a]
77123d6f	2024	FC[a]	77884a28	2024	ST[a]	7f6a881a	2024	PM
7ff45655	2024	FC[a]	832e667e	2024	FC[a]	89f778dc	2024	FC[a]
8d53b814	2024	FC[a]	91af3b10	2024	FC+MSR[a]	92657c9c	2024	FC[a]
98ac4866	2024	FC[a]	98edc5c2	2024	ST+MSR	9909f369	2024	FC[a]
9cfc7dc4	2024	FC[a]	9d044dea	2024	FC[a]	9ffa3123	2024	ST+FC+MSR
a1f34a25	2024	FC[a]	a3939160	2024	FC+MSR[a]	a621685d	2024	FC[a]
a694e76a	2024	FC[a]	a831d69b	2024	FC[a]	ab69dd0d	2024	FC[a]
ad7beca4	2024	FC[a]	b550575e	2024	FC[a]	b96c8b39	2024	FC+MSR[a]
bca443d4	2024	FC[a]	bcf3c5f4	2024	FC[a]	c7d555aa	2024	PM
c8c6d2de	2024	FC[a]	c92f7782	2024	FC[a]	c94e61ca	2024	FC+MSR[a]
c9b523c4	2024	FC[a]	ca97495a	2024	FC[a]	ce3171f1	2024	FC[a]
cf1e0ba4	2024	FC[a]	cf7f8d85	2024	FC[a]	d0fe9e8b	2024	PM+MSR[a]
d36b0adf	2024	PM[a]	d72e7ff3	2024	PM+MSR[a]	d85e760e	2024	FC+MSR[a]
db543f8e	2024	FC[a]	dd3b63a3	2024	MSR[a]	dfe6510c	2024	FC[a]
e14e8301	2024	FC[a]	e28b86a2	2024	FC+MSR[a]	ecdae240	2024	MSR[a]
ee93827a	2024	FC+MSR[a]	ef95373d	2024	FC[a]	efafc74a	2024	FC[a]
f126adac	2024	ST+MSR[a]	f1bea788	2024	FC[a]	f1c6b04e	2024	FC[a]
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References

[1] J. E. Cook, A. L. Wolf, Automating process discovery through event-data analysis, in: Proceedings of the 17th International Conference on Software Engineering (ICSE 1995), ACM, 1995, pp. 73–82. doi:10.1145/225014.225021.

[2] J. E. Cook, A. L. Wolf, Discovering models of software processes from event-based data, ACM Transactions on Software Engineering and Methodology 7 (3) (1998) 215–249. doi:10.1145/287000.287001.

[3] V. Rubin, C. W. Günther, W. M. P. van der Aalst, E. Kindler, B. F. van Dongen, W. Schäfer, Process mining framework for software processes, in: International Conference on Software Process (ICSP 2007), Vol. 4470 of LNCS, Springer, 2007, pp. 169–181. doi:10.1007/978-3-540-72426-1_15.

[4] J. A. Whittaker, M. G. Thomason, A Markov chain model for statistical software testing, IEEE Transactions on Software Engineering 20 (10) (1994) 812–824. doi:10.1109/32.328991.

[5] N. B. Ali, K. Petersen, C. Wohlin, A systematic literature review on the industrial use of software process simulation, Journal of Systems and Software 97 (2014) 65–85. doi:10.1016/j.jss.2014.06.059.

[6] H. Kagdi, M. L. Collard, J. I. Maletic, A survey and taxonomy of approaches for mining software repositories in the context of software evolution, Journal of Software Maintenance and Evolution: Research and Practice 19 (2) (2007) 77–131. doi:10.1002/smr.344.

[7] T. Hall, S. Beecham, D. Bowes, D. Gray, S. Counsell, A systematic literature review on fault prediction performance in software engineering, IEEE Transactions on Software Engineering 38 (6) (2012) 1276–1304. doi:10.1109/TSE.2011.103.

[8] A. E. Márquez-Chamorro, M. Resinas, A. Ruiz-Cortés, Predictive monitoring of business processes: A survey, IEEE Transactions on Services Computing 11 (6) (2018) 962–977. doi:10.1109/TSC.2017.2772256.

[9] I. Teinemaa, M. Dumas, M. La Rosa, F. M. Maggi, Outcome-oriented predictive process monitoring: Review and benchmark, ACM Transactions on Knowledge Discovery from Data 13 (2) (2019) 1–57. doi:10.1145/3301300.

[10] I. Verenich, M. Dumas, M. La Rosa, F. M. Maggi, I. Teinemaa, Survey and cross-benchmark comparison of remaining time prediction methods in business process monitoring, ACM Transactions on Intelligent Systems and Technology 10 (4) (2019) 1–34. doi:10.1145/3331449.

[11] E. Rama-Maneiro, J. C. Vidal, M. Lama, Deep learning for predictive business process monitoring: Review and benchmark, IEEE Transactions on Services Computing 16 (1) (2023) 739–756.

[12] C. d. S. Garcia, A. Meincheim, E. R. Faria Junior, M. R. Dallagassa, D. M. V. Sato, D. R. Carvalho, E. A. P. Santos, E. E. Scalabrin, Process mining techniques and applications – a systematic mapping study, Expert Systems with Applications 133 (2019) 260–295. doi:10.1016/j.eswa.2019.05.003.

[13] M. Thiede, D. Fuerstenau, A. P. Bezerra Barquet, How is process mining technology used by organizations? A systematic literature review of empirical studies, Business Process Management Journal 24 (4) (2018) 900–922. doi:10.1108/BPMJ-06-2017-0148.

[14] H. Zhang, B. Kitchenham, D. Pfahl, Software process simulation modeling: An extended systematic review, in: New Modeling Concepts for Today's Software Processes (ICSP 2010), LNCS, Springer, 2010, pp. 309–320. doi:10.1007/978-3-642-14347-2_27.

[15] P. Mukala, A. Cerone, F. Turini, Process mining event logs from FLOSS data: State of the art and perspectives, in: Software Engineering and Formal Methods (SEFM 2014 Collocated Workshops), LNCS, Springer, 2015, pp. 182–198. doi:10.1007/978-3-319-15201-1_12.

[16] R. Thamizharasan, K. Appavoo, A review on software process mining using Petri nets, Asian Journal of Applied Sciences 9 (3) (2016) 131–142. doi:10.3923/ajaps.2016.131.142.

[17] E. Stepanov, A. Mitsyuk, Continuous software monitoring backed by process mining: a systematic literature review, International Journal of Data Science and Analytics 22 (1) (2026). doi:10.1007/s41060-025-01000-w.

[18] K. Petersen, S. Vakkalanka, L. Kuzniarz, Guidelines for conducting systematic mapping studies in software engineering: An update, Information and Software Technology 64 (2015) 1–18. doi:10.1016/j.infsof.2015.03.007.

[19] B. A. Kitchenham, S. Charters, Guidelines for performing systematic literature reviews in software engineering, Tech. Rep. EBSE-2007-01, EBSE (2007).

[20] C. Wohlin, Guidelines for snowballing in systematic literature studies and a replication in software engineering, in: Proceedings of the 18th International Conference on Evaluation and Assessment in Software Engineering (EASE 2014), ACM, 2014, pp. 38:1–38:10. doi:10.1145/2601248.2601268.

[21] B. A. Kitchenham, L. Madeyski, D. Budgen, SEGRESS: Software engineering guidelines for REporting secondary studies, IEEE Transactions on Software Engineering 49 (3) (2023) 1273–1298. doi:10.1109/TSE.2022.3174092.

[22] R. Wieringa, N. Maiden, N. Mead, C. Rolland, Requirements engineering paper classification and evaluation criteria: a proposal and a discussion, Requirements Engineering 11 (1) (2006) 102–107.

[23] M. J. Page, J. E. McKenzie, P. M. Bossuyt, et al., The PRISMA 2020 statement: An updated guideline for reporting systematic reviews, BMJ 372 (2021) n71. doi:10.1136/bmj.n71.

[24] S. Jo, R. Kwon, G. Kwon, Exploring collaboration patterns in GitHub using discrete time Markov chain, in: 30th Asia-Pacific Software Engineering Conference (APSEC 2023), IEEE, 2023, pp. 645–646. doi:10.1109/APSEC60848.2023.00090.

[25] S. Jo, R. Kwon, G. Kwon, Probabilistic model checking GitHub repositories for software project analysis, Applied Sciences 14 (3) (2024) 1260. doi:10.3390/app14031260.

[26] M. Ortu, G. Destefanis, T. Hall, D. Bowes, Fault-insertion and fault-fixing behavioural patterns in Apache software foundation projects, Information and Software Technology 158 (2023) 107187. doi:10.1016/j.infsof.2023.107187.

[27] R. Joshi, N. Kahani, Comparative study of reinforcement learning in GitHub pull request outcome predictions, in: 2024 IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER), IEEE, 2024, pp. 489–500. doi:10.1109/SANER60148.2024.00057.

[28] C.-K. Jeon, N. Kim, H. P. In, Probabilistic approach to predicting risk in software projects using software repository data, International Journal of Software Engineering and Knowledge Engineering 25 (6) (2015) 1017–1032. doi:10.1142/S0218194015500151.

[29] L. Tian, A. Noore, Modeling distributed software defect removal effectiveness in the presence of code churn, Mathematical and Computer Modelling 41 (4–5) (2005) 379–389. doi:10.1016/j.mcm.2004.10.021.

[30] M. Tyagi, D. K. Tyagi, L. K. Tyagi, N. Tyagi, D. Kumar, A. Sikandar, Agile planning and tracking of software projects using the state-space approach, in: Proceedings of International Conference on Communication and Computational Technologies, CCIS, Springer, 2021, pp. 355–371. doi:10.1007/978-981-16-0404-1_26.

[31] H. Washizaki, K. Honda, Y. Fukazawa, Predicting release time for open source software based on the generalized software reliability model, in: 2015 Agile Conference, IEEE, 2015, pp. 76–81. doi:10.1109/Agile.2015.19.

[32] I. Massitela, J. Assunção, A. R. Santos, P. Fernandes, A structured stochastic model for software project estimation in Waterfall models, in: Proceedings of the 30th International Conference on Software Engineering and Knowledge Engineering (SEKE 2018), 2018. doi:10.18293/SEKE2018-033.

[33] W. B. Mesmia, M. Escheikh, K. Barkaoui, DevOps workflow verification and duration prediction using non-Markovian stochastic Petri nets, Journal of Software: Evolution and Process 33 (3) (2021) e2329. doi:10.1002/smr.2329.

[34] S. Bhadra, A stochastic Petri net model of continuous integration and continuous delivery, in: 2022 IEEE International Symposium on Software Reliability Engineering Workshops (ISSREW), IEEE, 2022, pp. 114–117. doi:10.1109/ISSREW55968.2022.00050.

[35] S. Bhadra, B. Das, P. Silva, V. Nagaraju, L. Fiondella, A stochastic Petri net model of continuous integration and continuous delivery, in: 2023 Annual Reliability and Maintainability Symposium (RAMS), IEEE, 2023, pp. 1–6. doi:10.1109/RAMS51473.2023.10088212.

[36] Y. Li, H. Zhang, L. Dong, B. Liu, J. Ma, Constructing a hybrid software process simulation model in practice: An exemplar from industry, in: 2020 IEEE/ACM International Conference on Software and System Processes (ICSSP), IEEE, 2020, pp. 135–144. doi:10.1145/3379177.3388906.

[37] M. I. Lunesu, R. Tonelli, L. Marchesi, M. Marchesi, Assessing the risk of software development in agile methodologies using simulation, IEEE Access 9 (2021) 134240–134258. doi:10.1109/ACCESS.2021.3115941.

[38] T. Magennis, The economic impact of software development process choice: Cycle-time analysis and Monte Carlo simulation results, in: 2015 48th Hawaii International Conference on System Sciences (HICSS), IEEE, 2015, pp. 5055–5064. doi:10.1109/HICSS.2015.599.

[39] M. Nafreen, M. Luperon, L. Fiondella, V. Nagaraju, Y. Shi, T. Wandji, Connecting software reliability growth models to software defect tracking, in: 2020 IEEE 31st International Symposium on Software Reliability Engineering (ISSRE), IEEE, 2020, pp. 138–147. doi:10.1109/ISSRE5003.2020.00022.

[40] E. Incerto, A. Vandin, S. S. Ahrabi, Stochastic conformance checking based on variable-length Markov chains, Information Systems 133 (2025) 102561. doi:10.1016/j.is.2025.102561.

[41] O. López-Pintado, M. Dumas, Discovery and simulation of business processes with probabilistic resource availability calendars, in: 2023 5th International Conference on Process Mining (ICPM), IEEE, 2023, pp. 1–8. doi:10.1109/ICPM60904.2023.10271965.

[42] P. Jalote, D. Kamma, Studying task processes for improving programmer productivity, IEEE Transactions on Software Engineering 47 (4) (2021) 801–817. doi:10.1109/TSE.2019.2904230.

[43] M. Gupta, Improving software maintenance using process mining and predictive analytics, in: 2017 IEEE International Conference on Software Maintenance and Evolution (ICSME), IEEE, 2017, pp. 681–686. doi:10.1109/ICSME.2017.39.

[44] M. Pourbafrani, S. Jiao, W. M. P. van der Aalst, SIMPT: Process improvement using interactive simulation of time-aware process trees, in: Research Challenges in Information Science (RCIS 2021), Vol. 415 of LNBIP, Springer, 2021, pp. 588–594. doi:10.1007/978-3-030-75018-3_40.

[45] A. Buliga, F. Meneghello, R. Graziosi, M. Ronzani, Enhanced what-if scenarios generation by bridging generative models and process simulation, in: 2025 7th International Conference on Process Mining (ICPM), IEEE, 2025, pp. 1–8. doi:10.1109/ICPM66919.2025.11220730.

[46] J. Caldeira, F. Brito e Abreu, J. Cardoso, J. Reis, Unveiling process insights from refactoring practices, Computer Standards & Interfaces 81 (2022) 103587. doi:10.1016/j.csi.2021.103587.

[47] M. A. Guinea-Cabrera, J. A. Holgado-Terriza, P. A. Pico-Valencia, PRIMAD: Building digital twins for software development lifecycle within an evolutionary architecture, in: 2025 IEEE Smart World Congress (SWC), IEEE, 2025, pp. 1680–1685. doi:10.1109/SWC65939.2025.00261.

[48] A. Burke, S. J. J. Leemans, M. T. Wynn, Stochastic process discovery by weight estimation, in: Process Mining Workshops (ICPM 2020), LNBIP, Springer, 2021.